Machine Intelligence 1

1.4 Additional Topics

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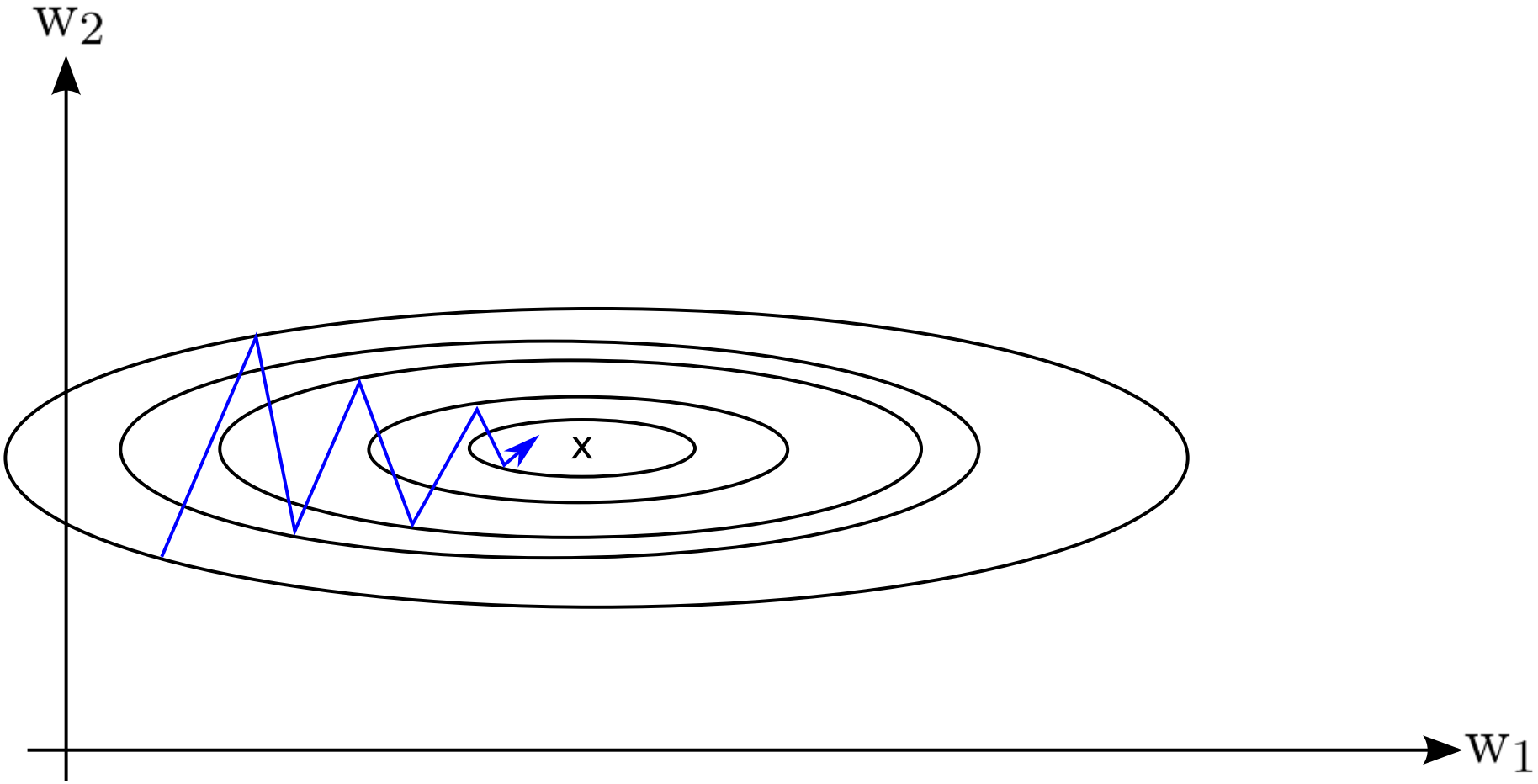
Fachgebiet Neuronale Informationsverarbeitung (NI)

WS 25/26

1.4.1 Improving Gradient Descent Optimization (Batch)

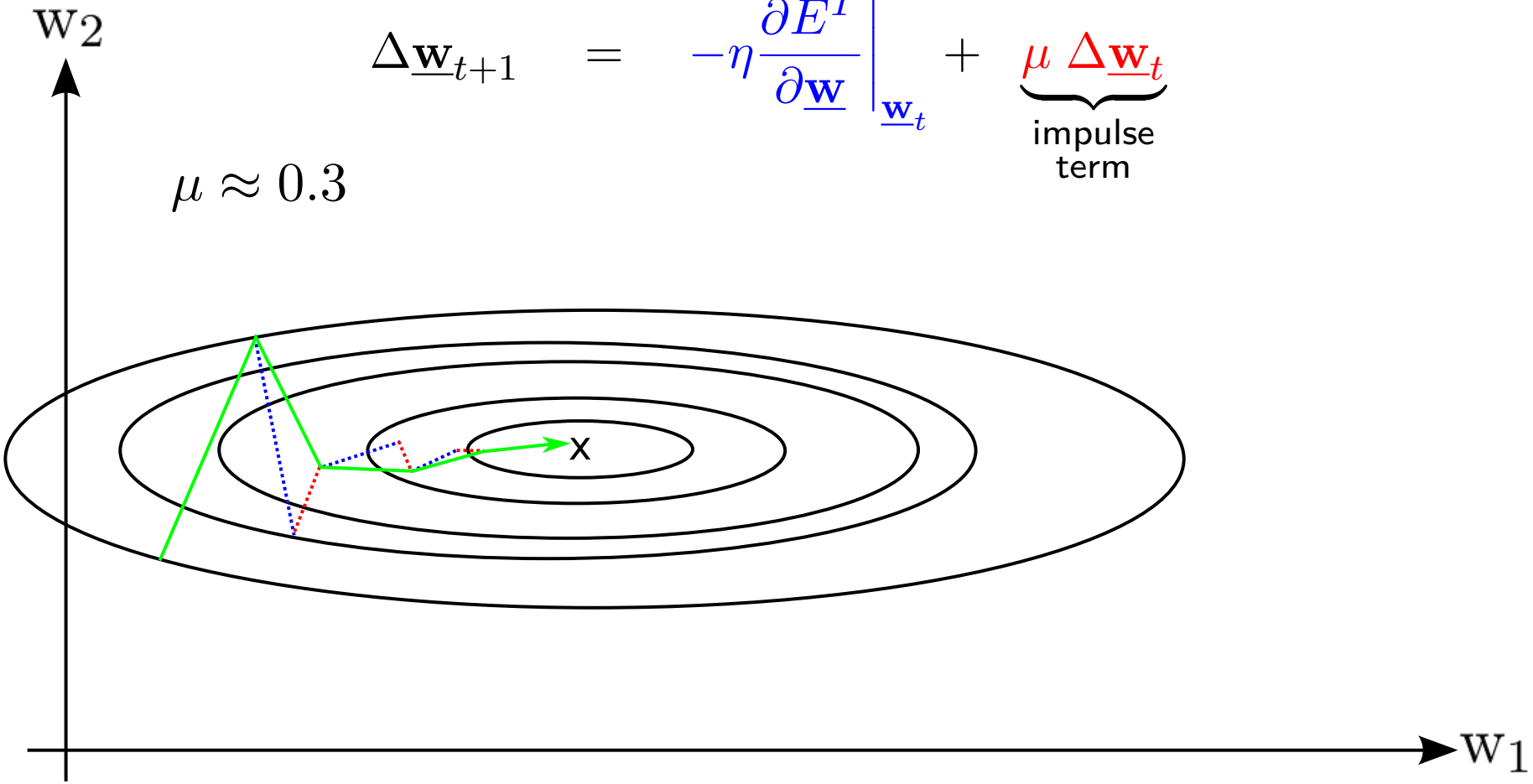
Impulse terms



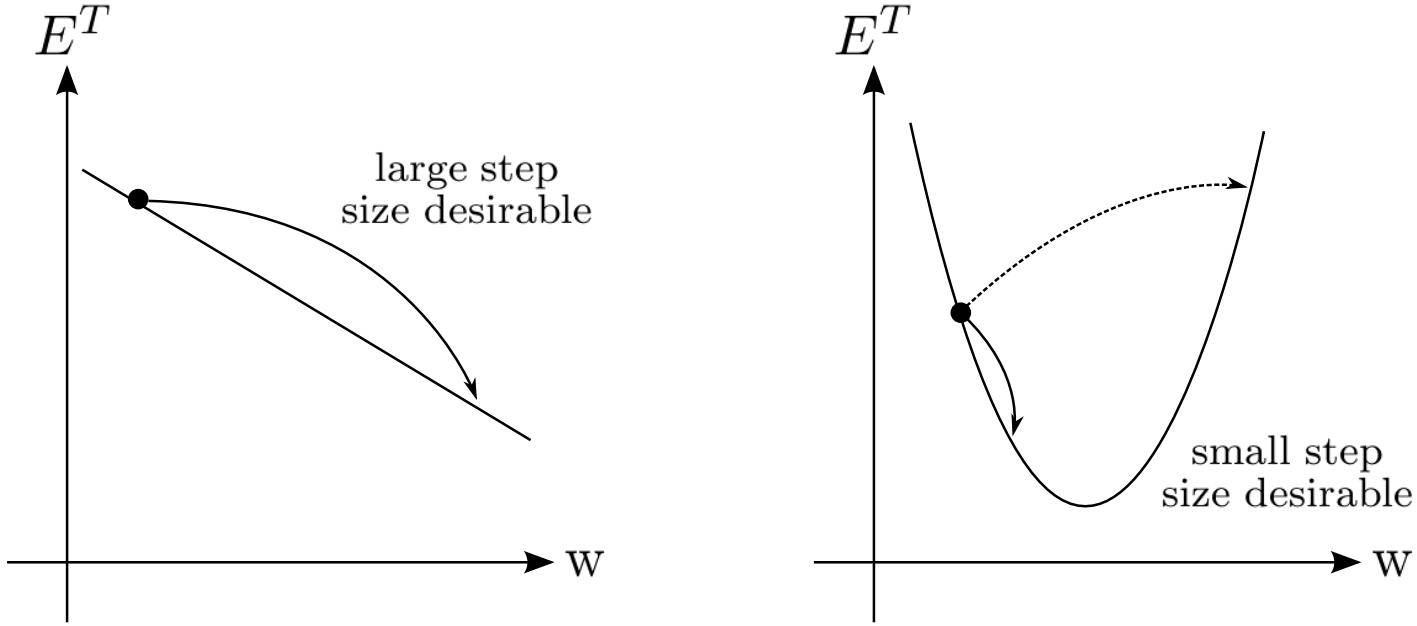
Impulse terms

$$\Delta \underline{\mathbf{w}}_{t+1} = -\eta \frac{\partial E^T}{\partial \underline{\mathbf{w}}}\bigg|_{\underline{\mathbf{w}}_t} + \underbrace{\mu\, \Delta \underline{\mathbf{w}}_t}_{\text{impulse term}}$$

$\mu \approx 0.3$



Adaptive step size

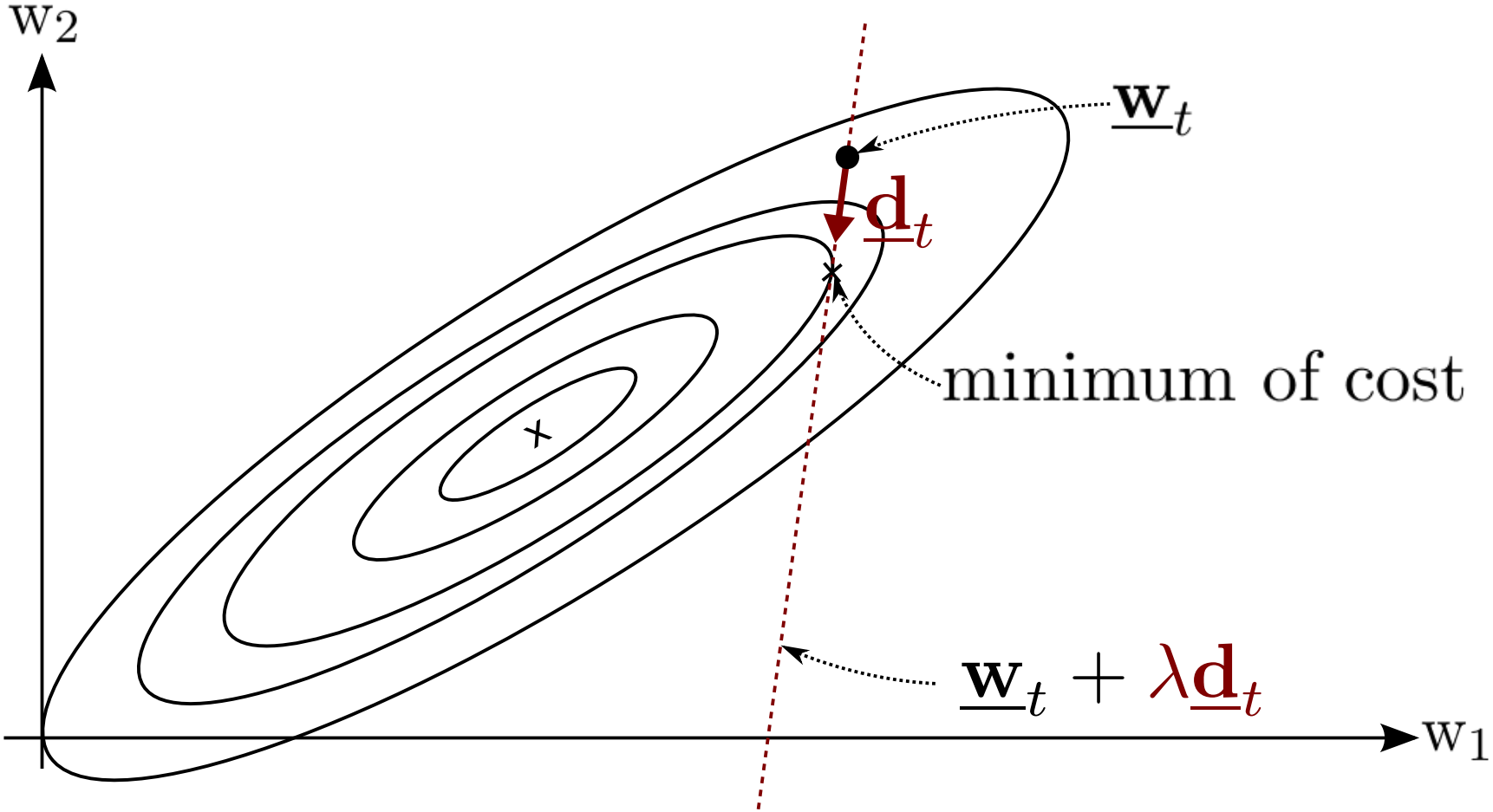


$$\eta_{t+1} = \begin{cases} \rho\eta_t, & \text{if } \Delta E^T < 0, \quad \text{increase step size, if } E^T \downarrow \\ \delta\eta_t, & \text{if } \Delta E^T > 0, \quad \text{decrease step size, if } E^T \uparrow \end{cases}$$

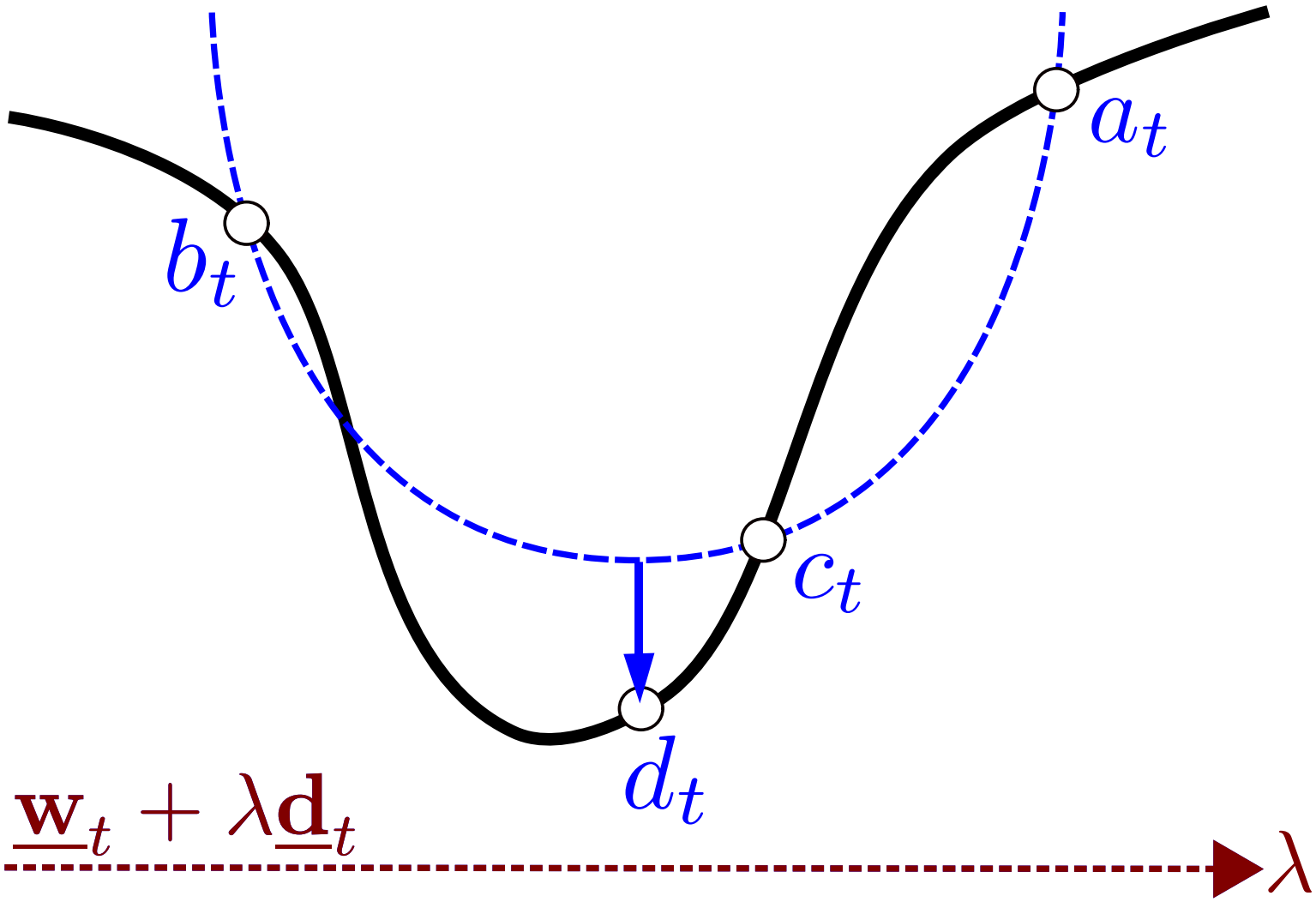
typical values: $\rho = 1.1, \delta = 0.5$

1.4.2 The Conjugate Gradient Method

Optimal step size



Parabolic interpolation



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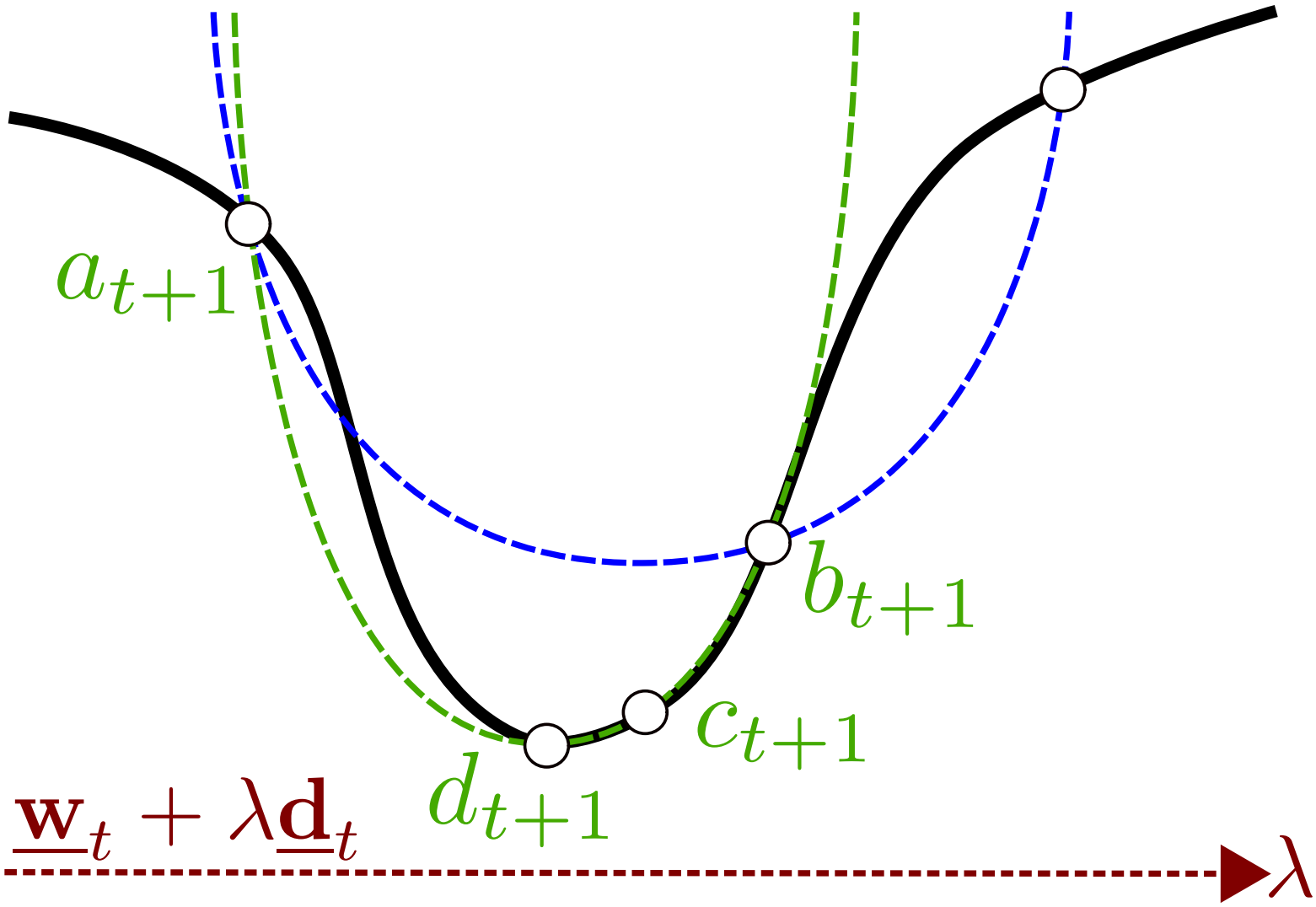


Line search

Line search: successive parabolic interpolation

Initialization: a_0, b_0, c_0 (on $\underline{\mathbf{w}}_t + \lambda \underline{\mathbf{d}}_t$) : $\quad E^T_{(a_0)}, \quad E^T_{(b_0)} > E^T_{(c_0)}$

while *stopping criterion not fulfilled* **do**

Fit a parabola through the three points a_t, b_t, c_t

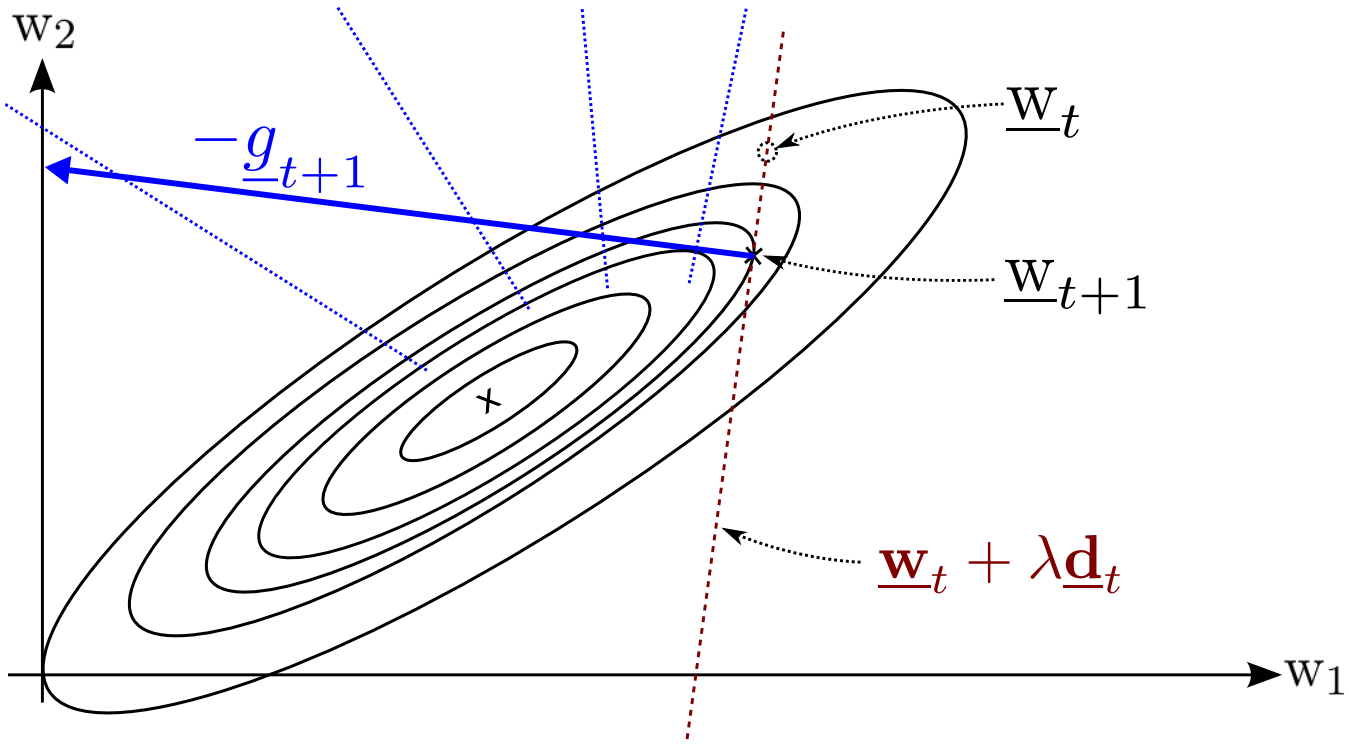
Calculate location d_t of its minimum

Set $c_{t+1} = d_t, \quad b_{t+1} = c_t, \quad a_{t+1} = \begin{cases} a_t, & E^T_{(a_t)} < E^T_{(b_t)} \\ b_t, & \text{else} \end{cases}$

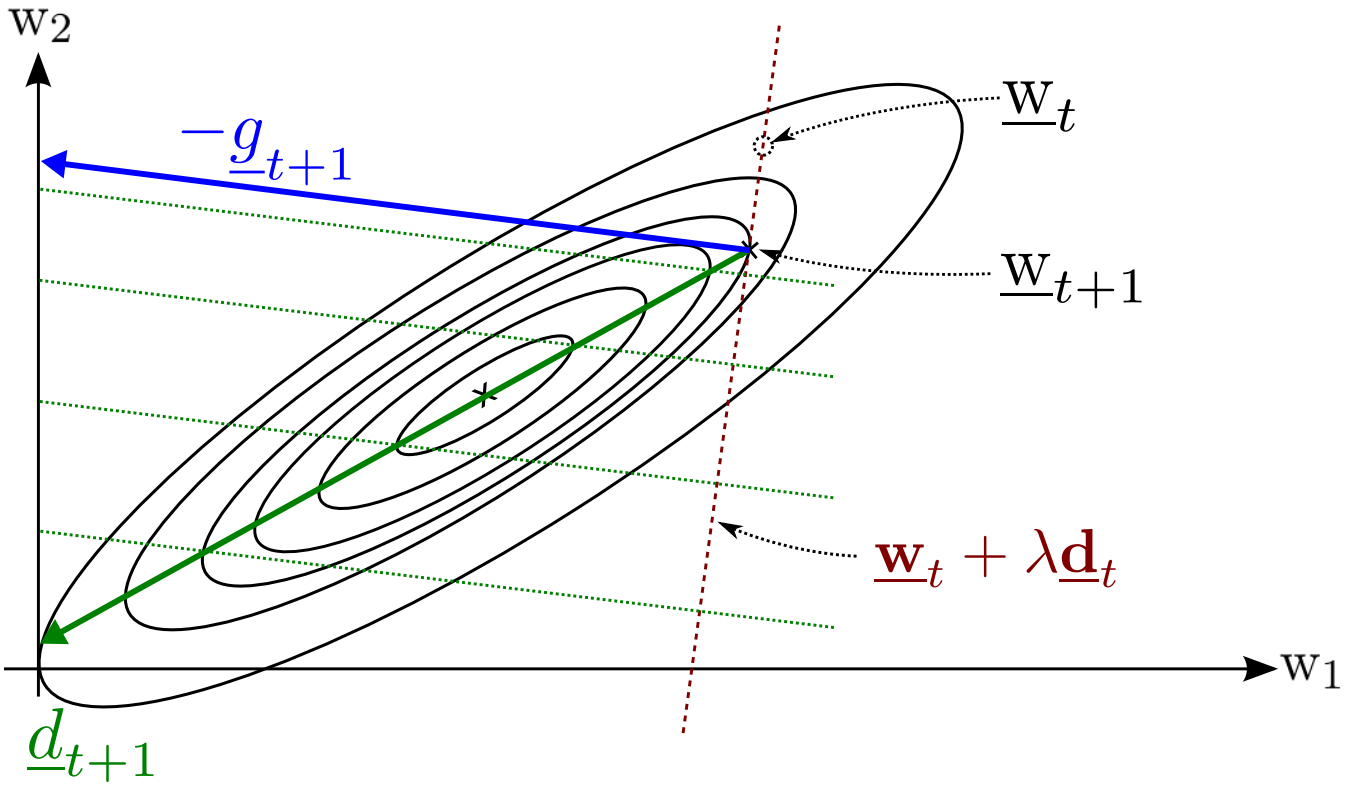
end

For details and implementation see, e.g., Press et al., 2007, Numerical Recipes, 2nd edition, Chapter 10.2.

The conjugate direction: parabolic cost functions



The conjugate direction: parabolic cost functions



$$\underline{\mathbf{d}}_t^\top \underline{\mathbf{H}}\, \underline{\mathbf{d}}_{t+1} \stackrel{!}{=} 0\,, \qquad H_{ij} := \frac{\partial^2 E^T}{\partial w_i \partial w_j} \quad \text{(Hessian Matrix)}$$

Adaptive momentum: the Polak-Ribiere rule

For general cost functions, we do not have the Hessian.

Therefore, we formulate the computation for parabolic cost functions in terms of gradients only.

$$\underline{\mathbf{g}}_{t+1} := \left.\frac{\partial E^T}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}_{t+1}} \qquad \text{(gradient at } \underline{\mathbf{w}}_{t+1}\text{)}$$

$$\underline{\mathbf{d}}_{t+1} = -\underline{\mathbf{g}}_{t+1} + \beta_t \underline{\mathbf{d}}_t \qquad \text{(conjugate direction)}$$

$$\beta_t = \underbrace{\frac{\underline{\mathbf{g}}_{t+1}^\top \left(\underline{\mathbf{g}}_{t+1} - \underline{\mathbf{g}}_t\right)}{\underline{\mathbf{g}}_t^\top \underline{\mathbf{g}}_t}}_{\text{Polak-Ribiere rule}} \qquad \text{(“smart momentum”)}$$

This formulation can now be generalized.

Conjugate gradient descent algorithm

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t + \eta_t \, \underline{\mathbf{d}}_t$$

Initialization: $\underline{\mathbf{w}}, \quad \underline{\mathbf{d}} = -\underline{\mathbf{g}}$

while *stopping criterion not fulfilled* **do**

Minimize E^T along $\underline{\mathbf{d}}$ using line-search $\rightarrow$ new $\underline{\mathbf{w}}$

Calculate the new conjugate direction, e.g., using the Polak-Ribiere rule for the calculation of the momentum.[1] $\rightarrow$ new $\underline{\mathbf{d}}$

end

- Subsequent search directions lose conjugacy
  - search direction should be reset to the steepest descent direction at least every N iterations

[1] Other formulas for calculating β exist.

1.4.3 Stochastic Approximation and Online Learning

Online learning

$$\Delta \mathrm{w}_{ij}^{v'v} \quad = \quad -\eta \frac{\partial E^T_{[\underline{\mathbf{w}}]}}{\partial \mathrm{w}_{ij}^{v'v}} \quad = \quad -\eta \frac{1}{p} \sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}_{[\underline{\mathbf{w}}]}}{\partial \mathrm{w}_{ij}^{v'v}}$$

Online learning

$$\Delta \mathrm{w}_{ij}^{v'v} \quad = \quad -\eta \frac{\partial E^T_{[\underline{\mathbf{w}}]}}{\partial \mathrm{w}_{ij}^{v'v}} \quad = \quad -\eta \frac{1}{p} \sum_{\alpha=1}^{p} \frac{\partial e^{(\alpha)}_{[\underline{\mathbf{w}}]}}{\partial \mathrm{w}_{ij}^{v'v}}$$

while *convergence criterion not met* **do**

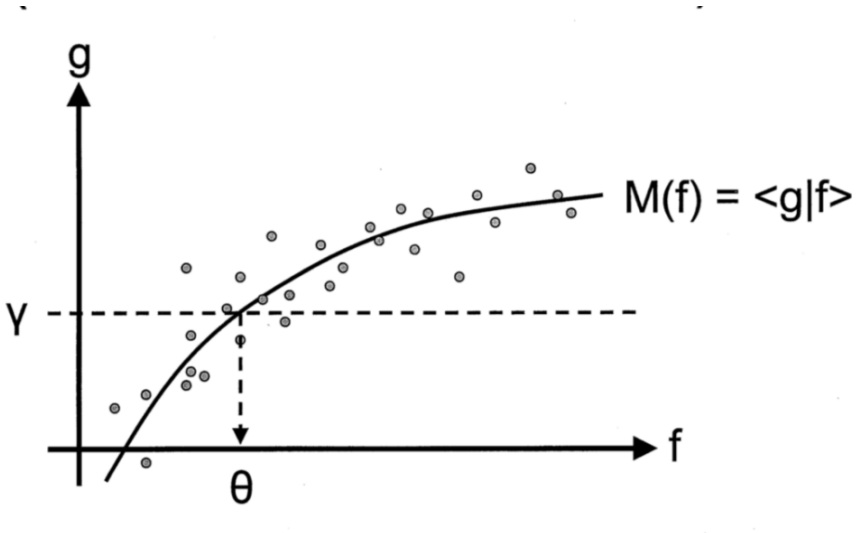
- select a data point: $\left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)}\right)$
- change weights according to: $\left(\Delta \mathrm{w}_{ij}^{v'v}\right)^{(t+1)} = -\eta_t \frac{\partial e^{(\alpha)}_{[\underline{\mathbf{w}}^{(t)}]}}{\partial \mathrm{w}_{ij}^{v'v}}$
- update learning rate η_t

end

On-line learning requires an iteration-dependent learning rate η_t to converge.

Justification (1)

Stochastic Approximation
Robbins and Monro (1951)



g, f: correlated random variables

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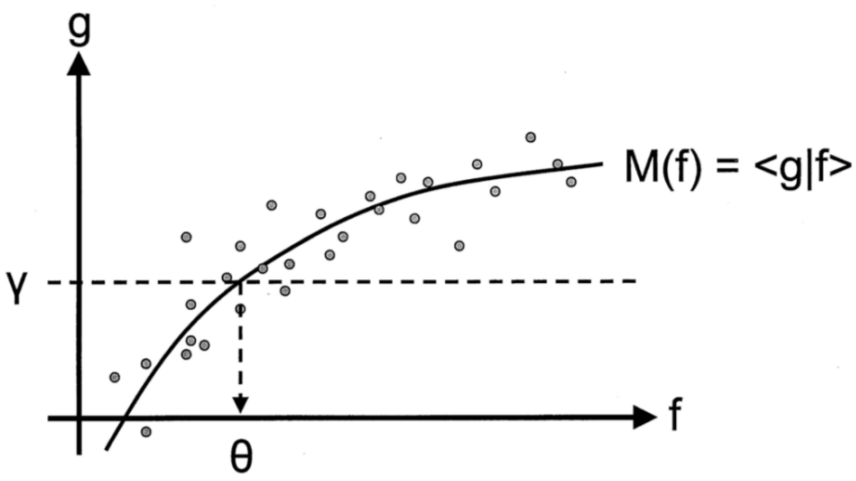


g, f: correlated random variables

Application to online learning:

- $f \rightarrow$ model parameters $\underline{\mathbf{w}}$
- $\underline{\mathbf{M}}(f) \rightarrow$ batch-gradient $\left.\frac{\partial E^G}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}}$
- $g \rightarrow$ individual gradients $\left.\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}}$
- $\theta \rightarrow$ optimal parameters $\underline{\mathbf{w}}^*$, i.e.

$$\left.\frac{\partial E^G}{\partial \underline{\mathbf{w}}}\right|_{\underline{\mathbf{w}}^*} = \underline{\mathbf{M}}(\theta) \stackrel{!}{=} \underline{\gamma} = \underline{\mathbf{0}}$$

Justification (2)

Let $g_t, f_t \in \mathbb{R}$, $g_t | f_t$, be correlated random variables and let the initial f_1 be an arbitrary real number. If

- g_t are bounded,
- $M(f)$ is monotonously increasing,
- $M(\theta) = \gamma$,
- and for the learning rates η_t,

$$\sum_{t=1}^{\infty} \eta_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty ,$$

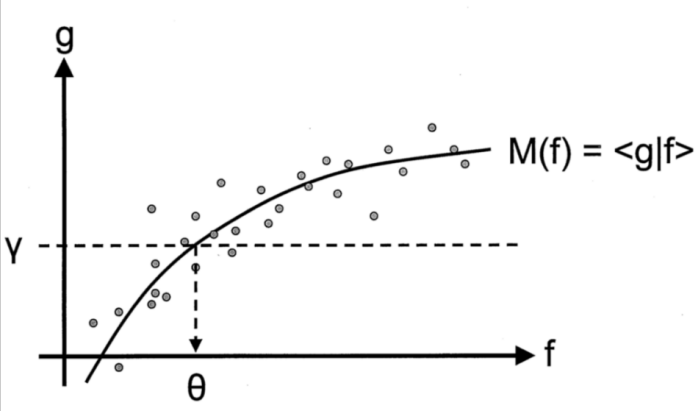
hold,

then the sequence

$$f_{t+1} = f_t + \eta_t(\gamma - g_t) , \qquad \eta_t > 0 ,$$

converges to θ in the sense $\lim_{t \to \infty} \langle (f_t - \theta)^2 \rangle = 0$.

Stochastic Approximation
Robbins and Monro (1951)



g, f: correlated random variables

Justification (2)

Let $\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}_t}, \underline{\mathbf{w}}_t \in \mathbb{R}^n$ be correlated random variables (n: number of weights)

and let the initial $\underline{\mathbf{w}}_1$ be an arbitrary real vector. If

- $\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}_t}$ are bounded,
- E^G is a strictly convex function, i.e., the Hessian $\underline{\mathbf{H}} = \left\{ \frac{\partial^2 E^G}{\partial w_i \partial w_j} \right\}$ is positive definite everywhere;
- $\frac{\partial E^G}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}^*} = 0$
- and for the learning rates η_t,

$$\sum_{t=1}^{\infty} \eta_t = \infty \quad \text{and} \quad \sum_{t=1}^{\infty} \eta_t^2 < \infty ,$$

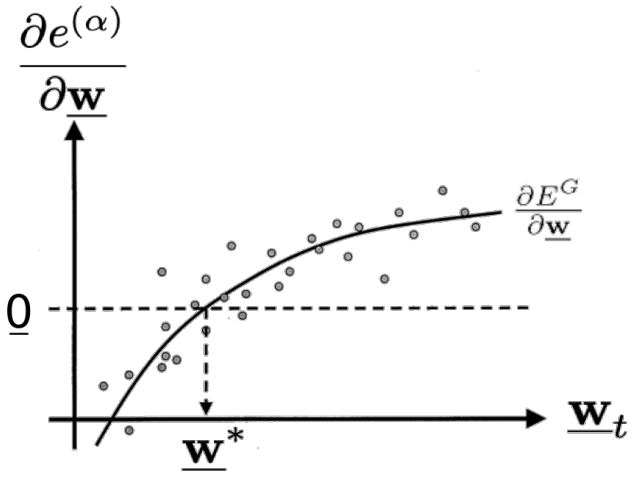
hold,

then the sequence

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \eta_t \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}\Big|_{\underline{\mathbf{w}}_t} , \qquad \eta_t > 0 ,$$

converges to $\underline{\mathbf{w}}^*$ in the sense $\lim_{t\to\infty} \langle (\underline{\mathbf{w}}_t - \underline{\mathbf{w}}^*)^2 \rangle = 0$.

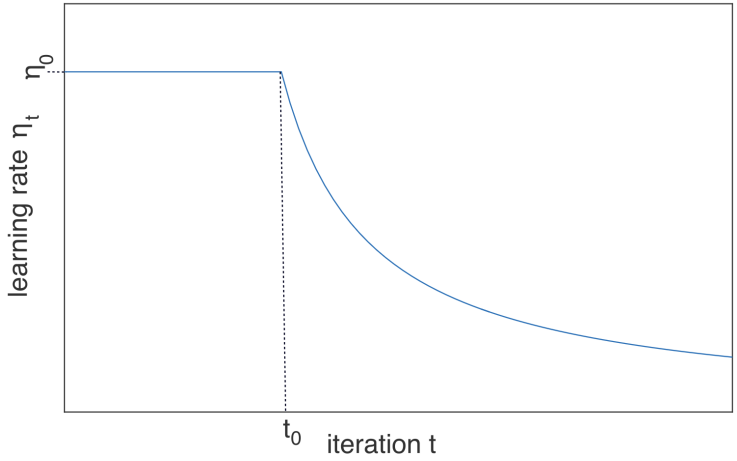
Stochastic Approximation
Robbins and Monro (1951)



Annealing of the learning rate η

- Asymptotically optimal learning rate: $\eta_t \propto \frac{1}{t}$.[2]
- Baseline annealing schedule:

$$\eta_t = \begin{cases} \eta_0 & \text{if} \quad 0 < t \leq t_0 \\ \frac{\eta_0}{t - t_0} & \text{if} \quad t > t_0 \end{cases}$$



- The hyperparameters η_0 and t_0 may be difficult to choose in practice.
- The algorithm is sensitive to the step size sequence, and using the asymptotically optimal learning rate too early can be quite harmful.

[2] Note that the theoretical results hold for the convex case only.

1.4.4 Improving Stochastic Gradient Descent

Adaptive Moment Estimation (ADAM)

- ADAM[3] is based on ideas related to impulse terms and adaptive step size.
- The noisy individual gradients $g_k^{(t)}$ are replaced by their running average $m_k^{(t)}$,

$$m_k^{(t)} = \beta_1 m_k^{(t-1)} + (1-\beta_1) g_k^{(t)},$$

  with, e.g., $g_k^{(t)} = \frac{\partial e^{(\alpha_t)}}{\partial w_k}$.

- Learning steps are scaled by a running average $(\sigma_k^{(t)})^2$ of the uncentered variance of the individual gradients,

$$(\sigma_k^{(t)})^2 = \beta_2 (\sigma_k^{(t-1)})^2 + (1-\beta_2)(g_k^{(t)})^2.$$

- The learning rule thus reads:[4] $$\Delta w_k^{(t)} = -\frac{\eta_t}{\sigma_k^{(t)} + \epsilon} m_k^{(t)}.$$

  The effective step size is reduced, if the variance increases, i.e., if the running estimates $m_k^{(t)}$ of the batch gradient become less reliable.

[3] Kingma and Ba (2015).

[4] A finite value of ϵ ($\epsilon > 0$) ensures that the denominator is always non-zero.

Removing the initialization bias

- $m_k^{(0)}$ and $\sigma_k^{(0)}$ are initialized with zero. $\Rightarrow$ The estimates $m_k^{(t)}$ and $\sigma_k^{(t)}$ are biased towards zero for small t.
- The bias can be corrected using the modified learning rule

$$\Delta w_k^{(t)} = -\eta_t \frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t} \frac{m_k^{(t)}}{\sigma_k^{(t)} + \epsilon}.$$

- Default settings for the hyperparameters:

$$\eta_t = \eta = 0.001,\ \beta_1 = 0.9,\ \beta_2 = 0.999,\ \epsilon = 10^{-8}.$$

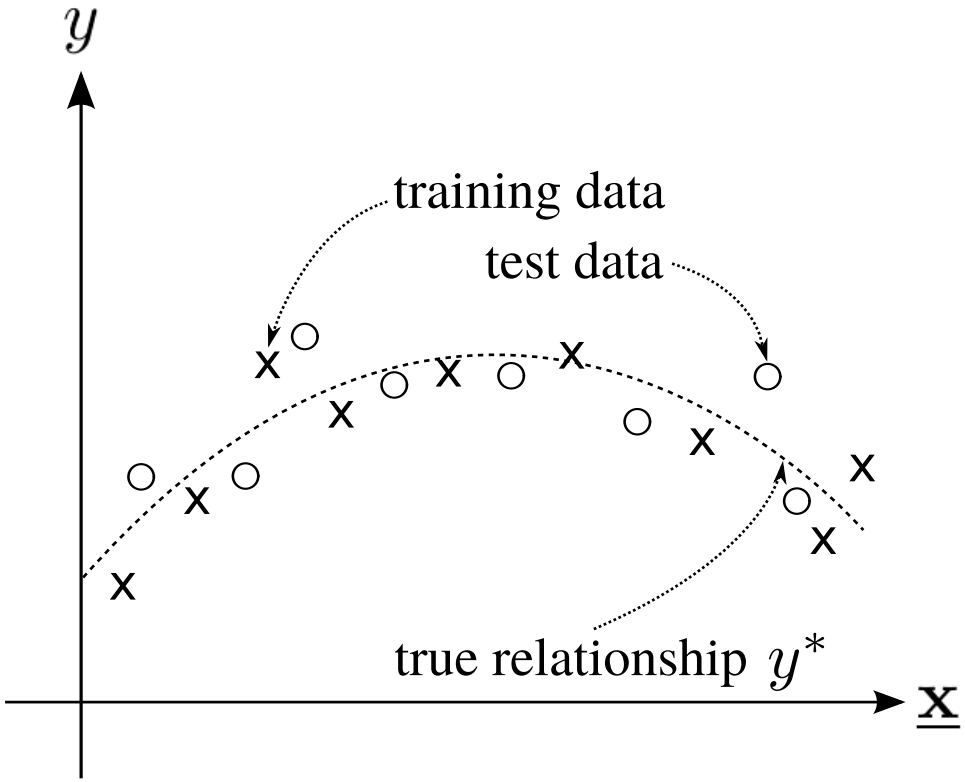
Annealing of the learning rate η and convergence

- Although a constant value $\eta_t = \eta = \text{const}_{(t)}$ may suffice for some optimization problems, others may require the annealing of the learning rate for convergence. Often $\eta_t \propto \frac{1}{\sqrt{t}}$ is used in practice.
- However, non-convergence proofs for ADAM exist even for an annealed learning rate $\eta_t \propto \frac{1}{\sqrt{t}}$ in the convex setting.[5]
- Convergence problems can be alleviated by a careful choice of the other hyperparameters ϵ, β_1, and β_2. Also variants of ADAM may be considered.

[5] Reddi et al. (2018).

1.4.5 Overfitting and Underfitting

Overfitting and underfitting



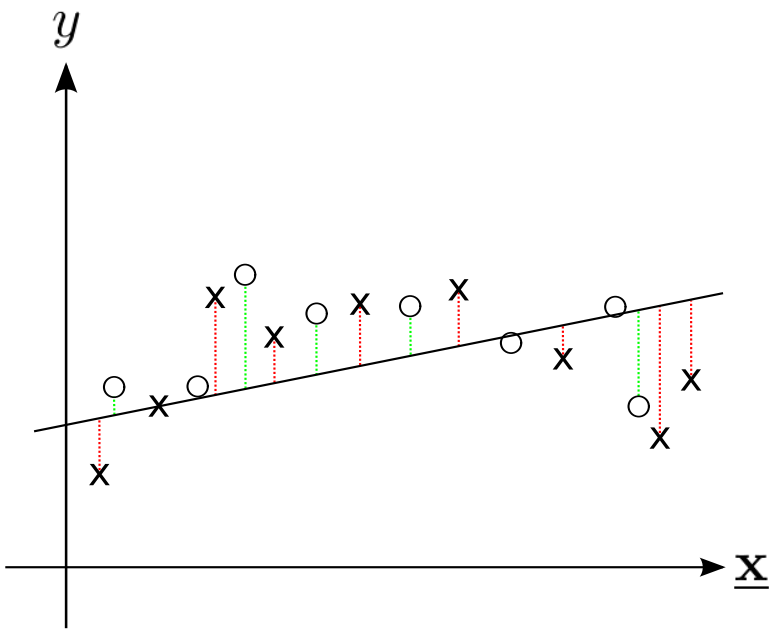
data generation:

e.g. $y_T = y^*_{(\underline{\mathbf{x}})} + \underbrace{\eta}_{\text{noise, zero mean}}$

Goal:

Find a **good** model for $y^*_{(\underline{\mathbf{x}})}$ (explanation, prediction)

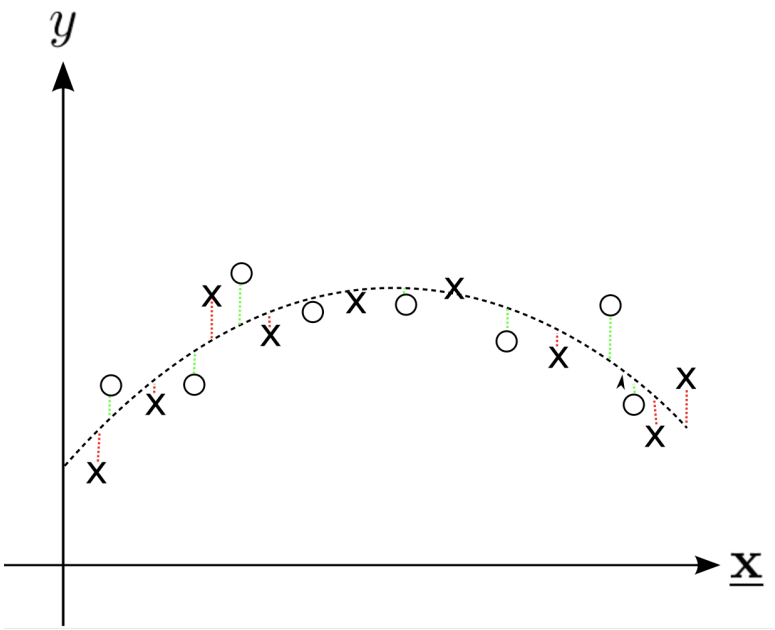
Underfitting



Diagnostics

$$\left.\begin{array}{l} E^T \text{ large} \\ E^G \text{ large} \end{array}\right\} E^T \approx E^G$$

Well fitted model



Diagnostics

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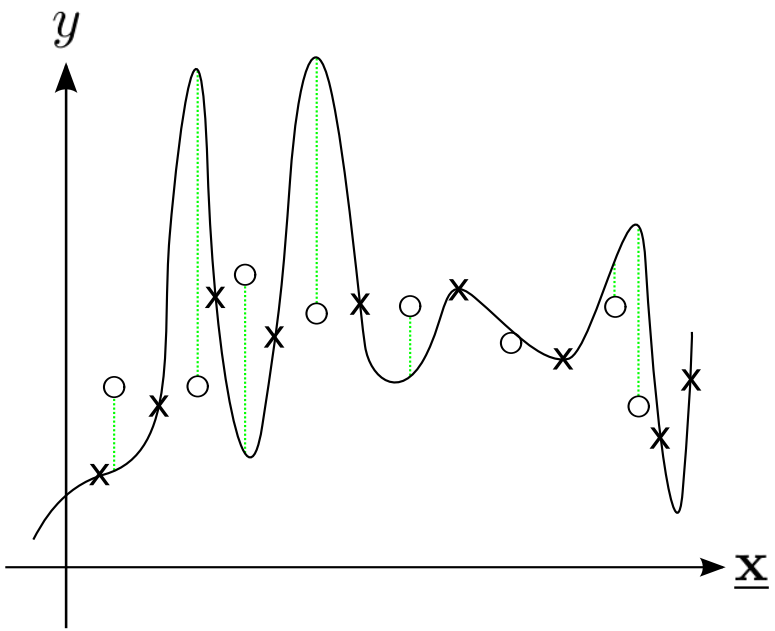
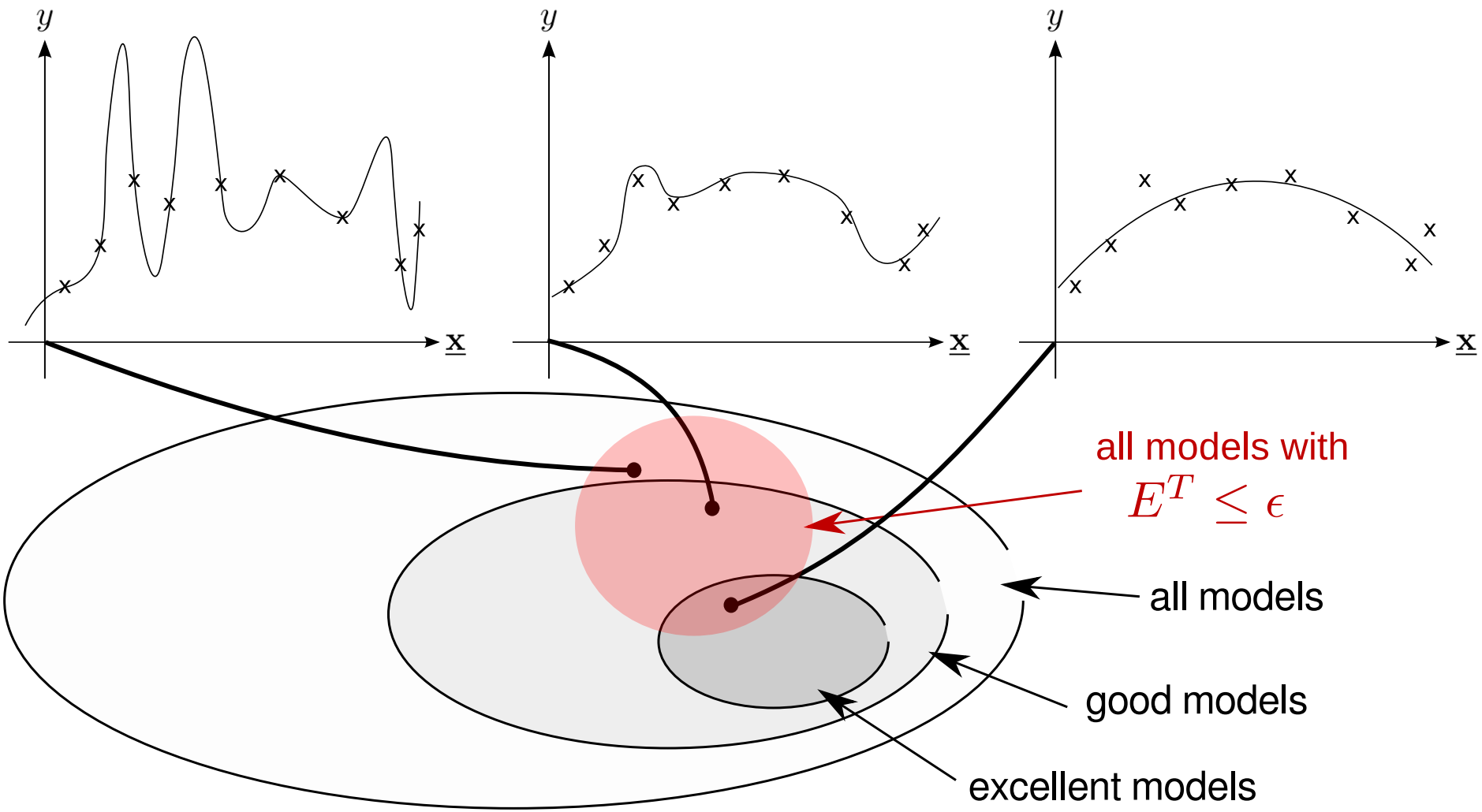
Overfitting



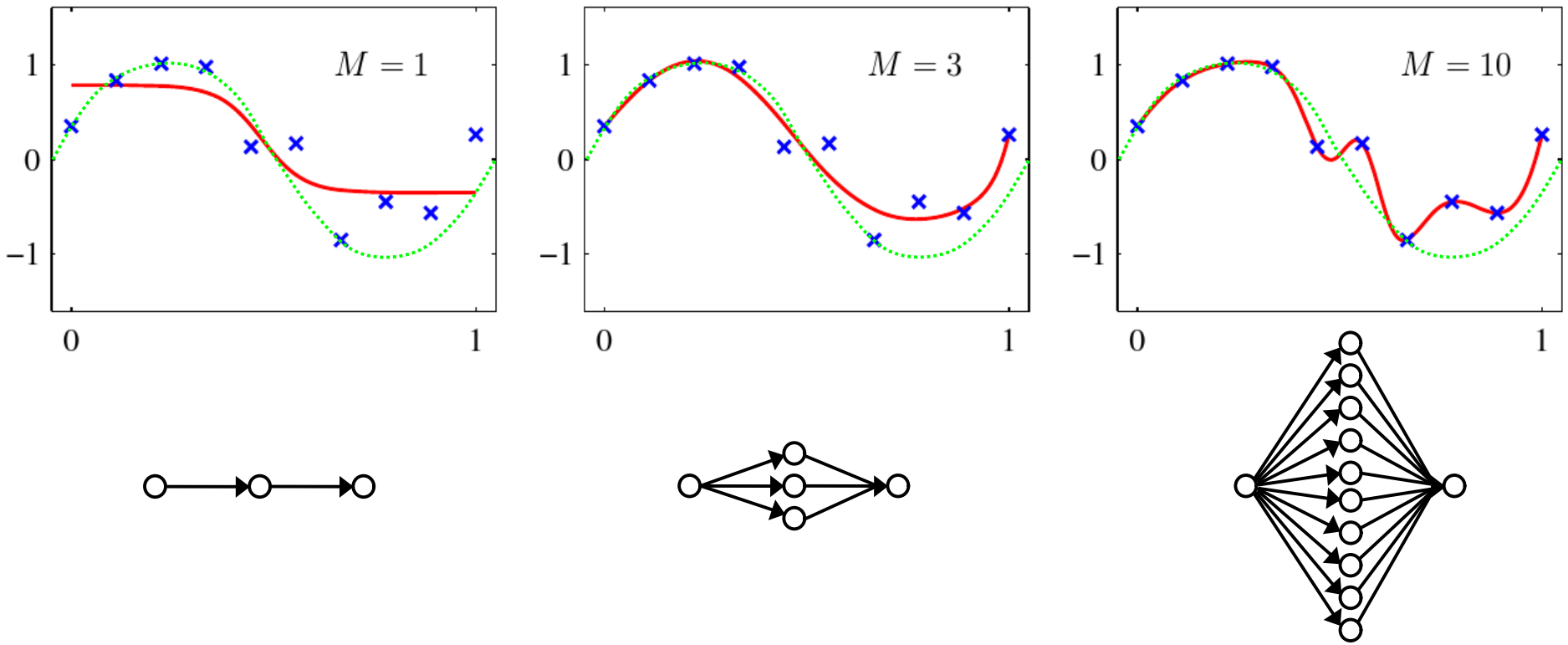
Diagnostics

$$\left.\begin{array}{l} E^T \text{ small} \\ E^G \text{ large} \end{array}\right\} E^T \ll E^G$$

Overfitting and model selection



Example with MLPs



MLP with M sigmoid hidden neurons $f_j^1(x) = \tanh(x)$ and linear output neuron $f_1^2(x) = x$, from Bishop (2006)

1.4.6 Bias and Variance

A learning problem

Example scenario

Observations:

$$y_T = \underbrace{y^*_{(\underline{\mathbf{x}})}}_{\substack{\text{true}\\ \text{relationship}}} + \underbrace{\eta}_{\text{noise}}, \quad \underline{\mathbf{x}} \in \mathbb{R}^N, \;\; y_T \in \mathbb{R}, \;\; \eta \sim \mathcal{N}(0, \sigma_\eta)$$

- $\underline{\mathbf{x}}, y_T$ are random variables.
- Many datasets of equal length are drawn iid from $P_{(y_T, \underline{\mathbf{x}})} = P_{(y_T|\underline{\mathbf{x}})} P_{(\underline{\mathbf{x}})}$.

A learning problem

Example scenario

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- $\underline{\mathbf{x}}, y_T$ are random variables.
- Many datasets of equal length are drawn iid from $P_{(y_T, \underline{\mathbf{x}})} = P_{(y_T|\underline{\mathbf{x}})} P_{(\underline{\mathbf{x}})}$.

- Fitting one MLP to every dataset:

$\rightarrow$ $\underline{\mathbf{w}}$ (model parameters) are random variables.

$\rightarrow$ $y_{(\underline{\mathbf{x}};\underline{\mathbf{w}})}$ (predicted values) are random variables.

Generalization error

Example: Quadratic cost

$$E^G = \langle e^G \rangle_{P_{(\underline{\mathbf{x}})}}, \text{ where } e^G = \langle (y_{(\underline{\mathbf{x}};\underline{\mathbf{w}})} - y_T)^2 \rangle_{P_{(y_T|\underline{\mathbf{x}})}}$$

$$\begin{aligned} e^G &= \langle (y - y_T)^2 \rangle \\ &= \langle (y - y^* + y^* - y_T)^2 \rangle \\ &= \langle (y - y^*)^2 \rangle + \langle (y^* - y_T)^2 \rangle + 2(y - y^*) \underbrace{\langle (y^* - y_T) \rangle}_{=0} \\ &= \langle (y - y^*)^2 \rangle + \sigma_\eta^2 \end{aligned}$$

$$\langle e^G \rangle_{\text{all datasets}} = \langle (y - y^*)^2 \rangle_{\text{all datasets}} + \sigma_\eta^2$$

Decomposition of the generalization error

Generalization error:

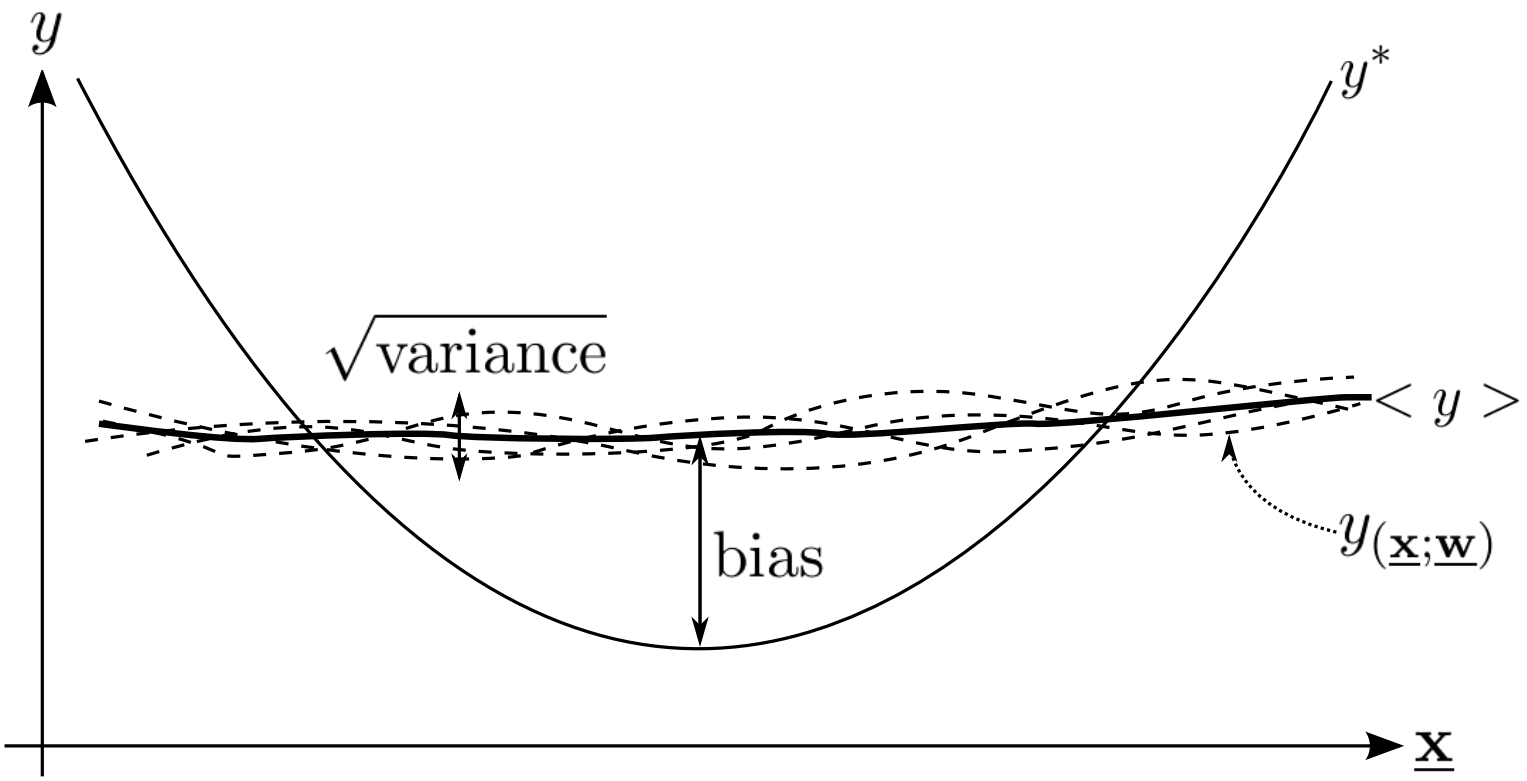
$$\langle e^G \rangle_{\text{all datasets}} = \langle \underbrace{(y - y^*)^2}_{\text{bias}^2 + \text{variance}} \rangle_{\text{all datasets}} + \sigma_\eta^2$$

Decomposing the first term we obtain:

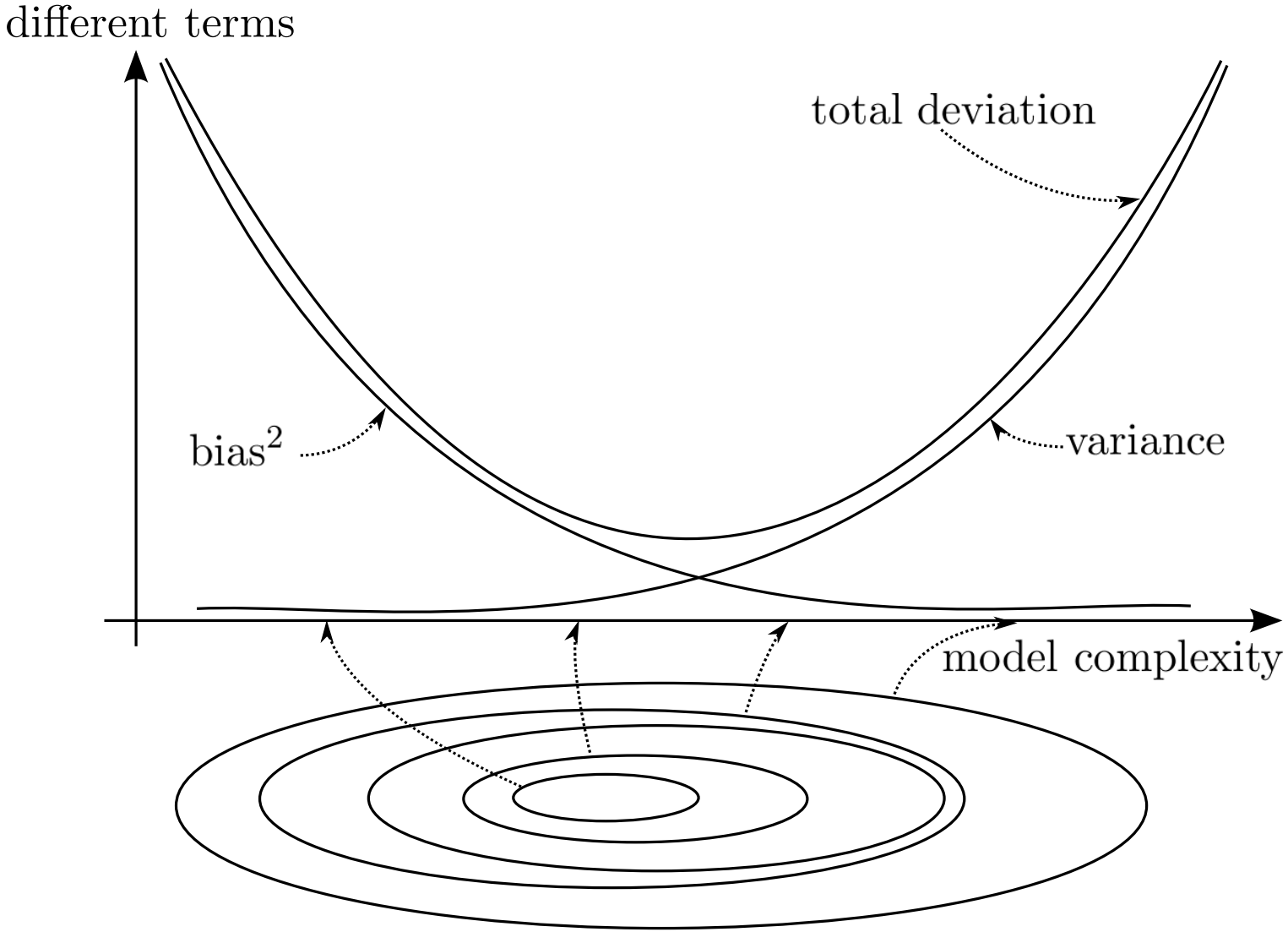
$$\begin{aligned} \langle (y - y^*)^2 \rangle &= \langle (y - \langle y \rangle + \langle y \rangle - y^*)^2 \rangle \\ &= \underbrace{(\langle y \rangle - y^*)^2}_{\text{bias}^2} + \underbrace{\langle (y - \langle y \rangle)^2 \rangle}_{\text{variance}} + 2 \underbrace{\langle y - \langle y \rangle \rangle}_{=0} (\langle y \rangle - y^*) \end{aligned}$$

The expectations are a function of $\underline{\mathbf{x}}$.

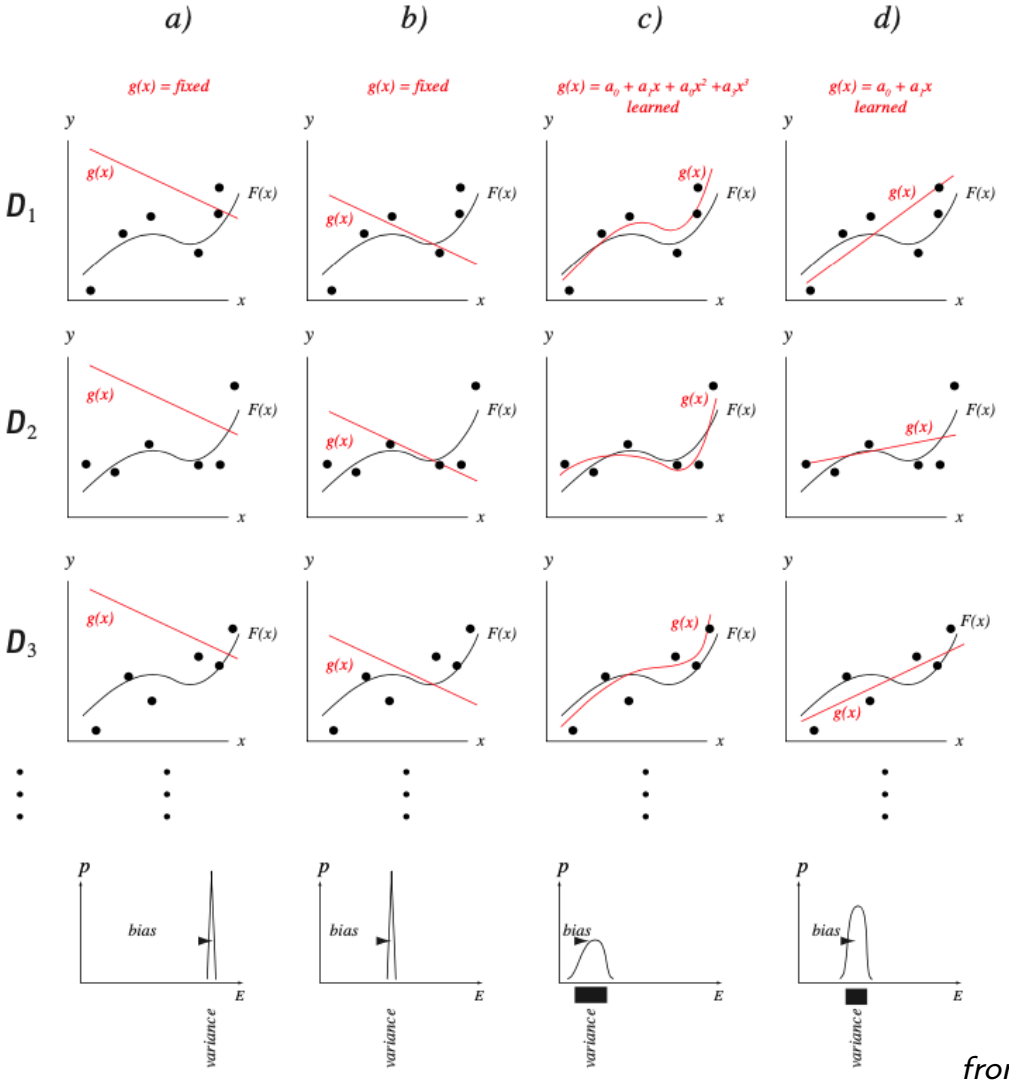
Bias and variance



Changes with model complexity



Regression example



from Hart et al., 2000, p. 467

$\Rightarrow$ Bias-variance trade-off applies to wide range of inductive learning problems.

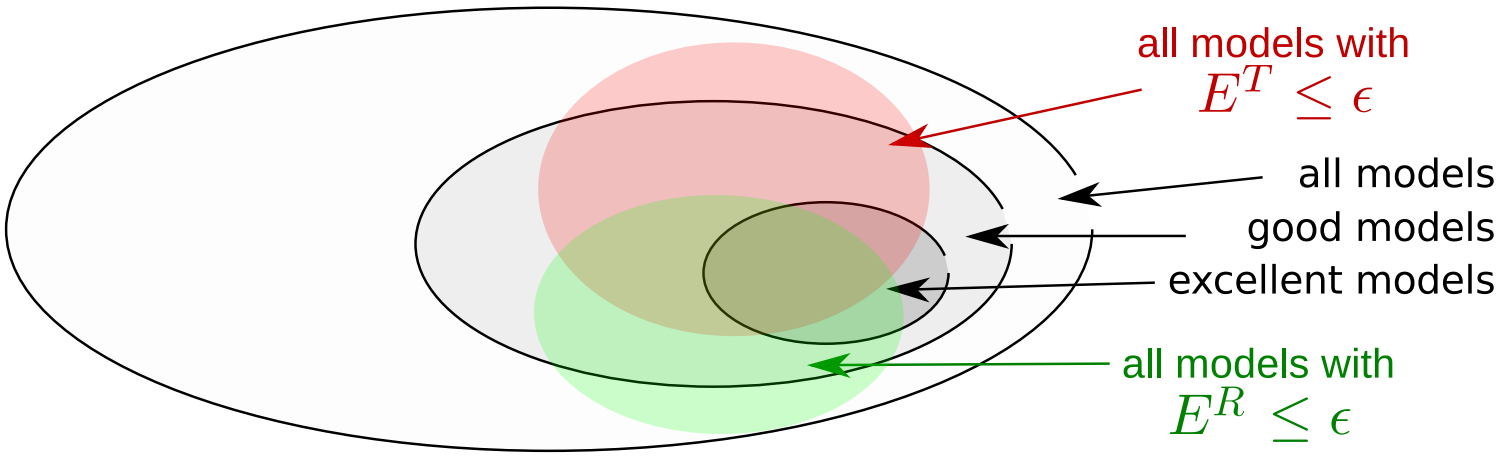
1.4.7 Regularization

Regularization

Risk function

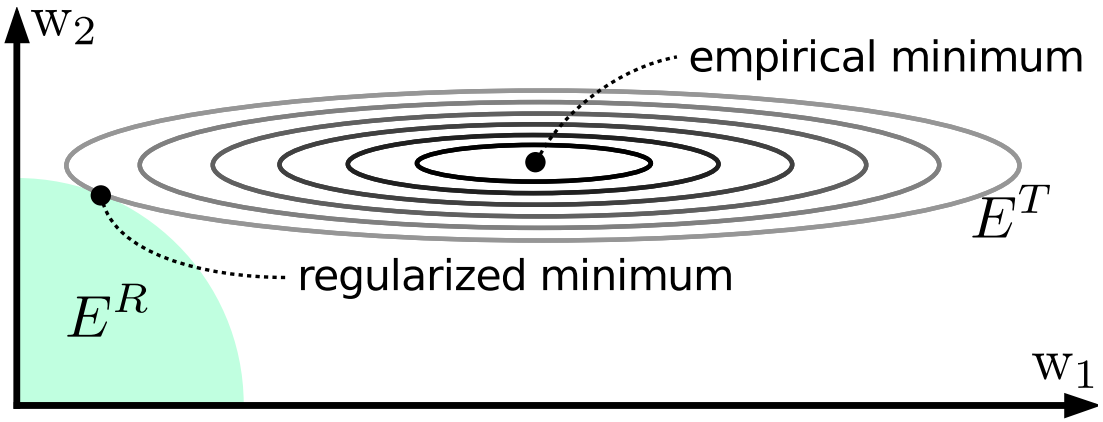
$$R_{[\underline{\mathbf{w}}]} = \underbrace{E^T_{[\underline{\mathbf{w}}]}}_{\substack{\text{training}\\ \text{error}}} + \underbrace{\lambda E^R_{[\underline{\mathbf{w}}]}}_{\substack{\text{regularization}\\ \text{term}}} \stackrel{!}{=} \min$$

- E^R : prior knowledge about the solution
- λ : regularization parameter



Regularization example: weight decay

$$E^R_{[\underline{\mathbf{w}}]} = \frac{1}{2} \sum_{(i,j,v',v)} \left(\mathrm{w}_{ij}^{v'v}\right)^2$$

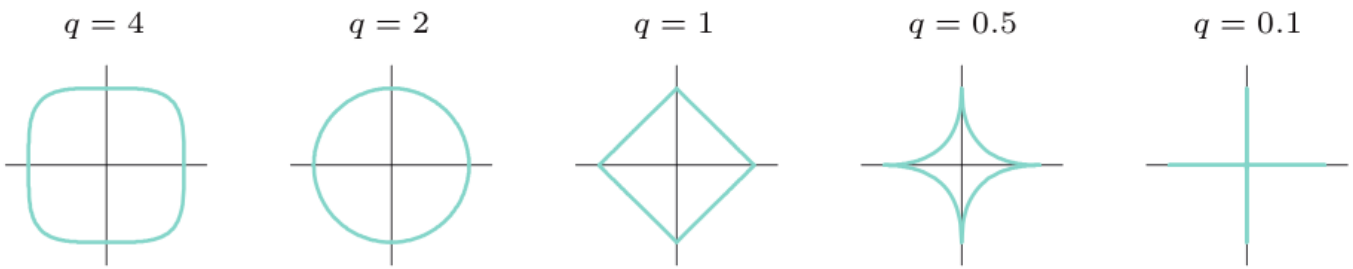


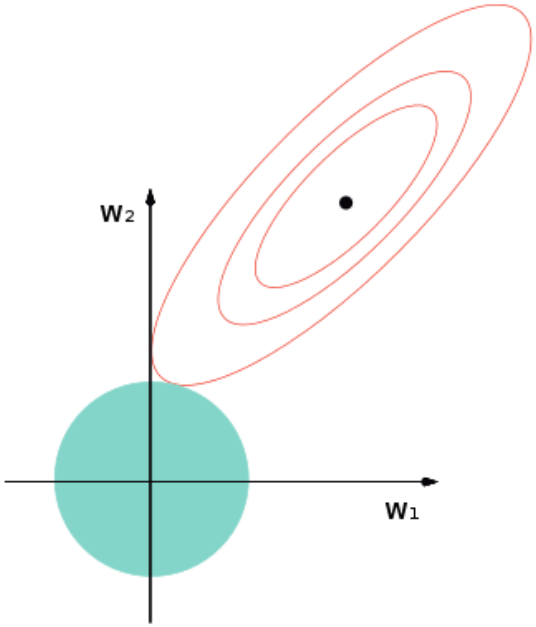
Minimization of R through gradient descent

$$\Delta \mathrm{w}_{ij}^{v'v} \sim -\frac{\partial R}{\partial \mathrm{w}_{ij}^{v'v}} = \underbrace{-\frac{\partial E^T}{\partial \mathrm{w}_{ij}^{v'v}}}_{\text{e.g. via backprop}} \underbrace{- \lambda \mathrm{w}_{ij}^{v'v}}_{\text{decay term}}$$

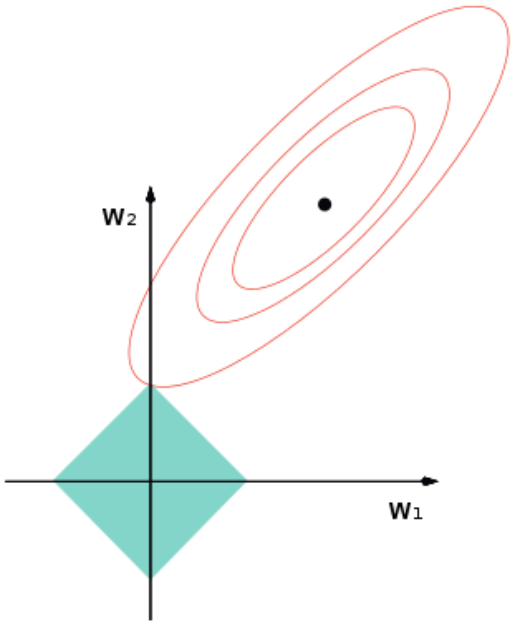
Other forms of weight decay type regularization

- general form of regularization: $E^R = \sum_{(i,j,v',v)} |w_{ij}^{v'v}|^q$





L_2: *weight decay* with $q = 2$



L_1: *lasso* with $q = 1$

Regularization example: symmetries

Odd vs. even function

$$E^R_{[\mathrm{w}]} = \frac{1}{2p} \sum_{\alpha=1}^{p} \left(y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \pm y_{(-\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right)^2$$

Regularization example: symmetries

Odd vs. even function

$$E^R_{[\mathrm{w}]} = \frac{1}{2p} \sum_{\alpha=1}^{p} \left(y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \pm y_{(-\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right)^2$$

Invariance under translation:

$$E^R_{[\mathrm{w}]} = \frac{1}{2p} \sum_{\alpha=1}^{p} \left(y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} - y_{(\underline{\mathbf{x}}^{(\alpha)}-\underline{\mathbf{t}};\underline{\mathbf{w}})} \right)^2$$

Regularization example: symmetries

Odd vs. even function

$$E^R_{[\mathrm{w}]} = \frac{1}{2p} \sum_{\alpha=1}^{p} \left(y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \pm y_{(-\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right)^2$$

Invariance under translation:

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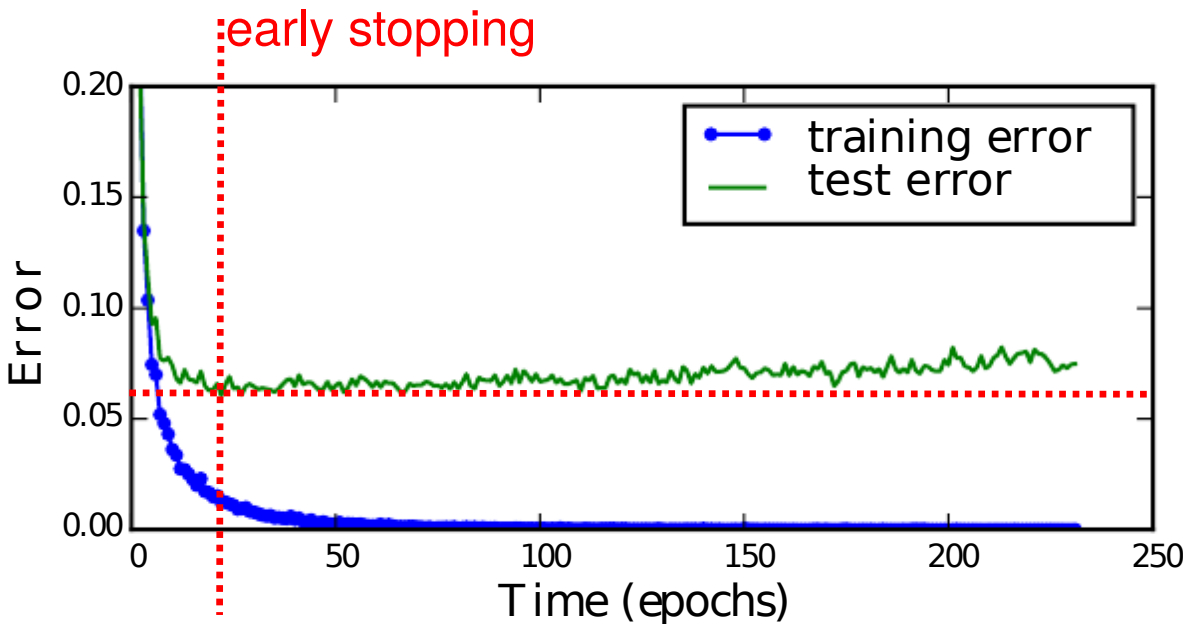
Monotony:

$$E^R_{[\mathrm{w}]} = \frac{1}{n_p} \sum_{\text{“}\underline{\mathbf{x}}^{(\alpha)} > \underline{\mathbf{x}}^{(\beta)}\text{”}} \begin{cases} \left(y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} - y_{(\underline{\mathbf{x}}^{(\beta)};\underline{\mathbf{w}})} \right)^2 & \text{if } y_{(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} < y_{(\underline{\mathbf{x}}^{(\beta)};\underline{\mathbf{w}})} \\ 0 & \text{else} \end{cases}$$

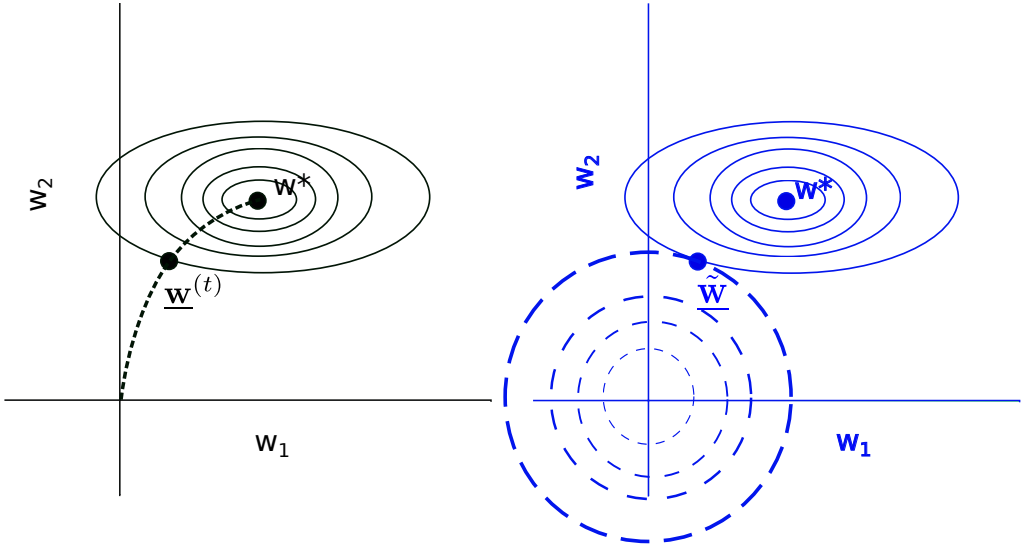
Early stopping



- Estimate generalization error with a test set during training.
- Stop training when the error on the test set rises.

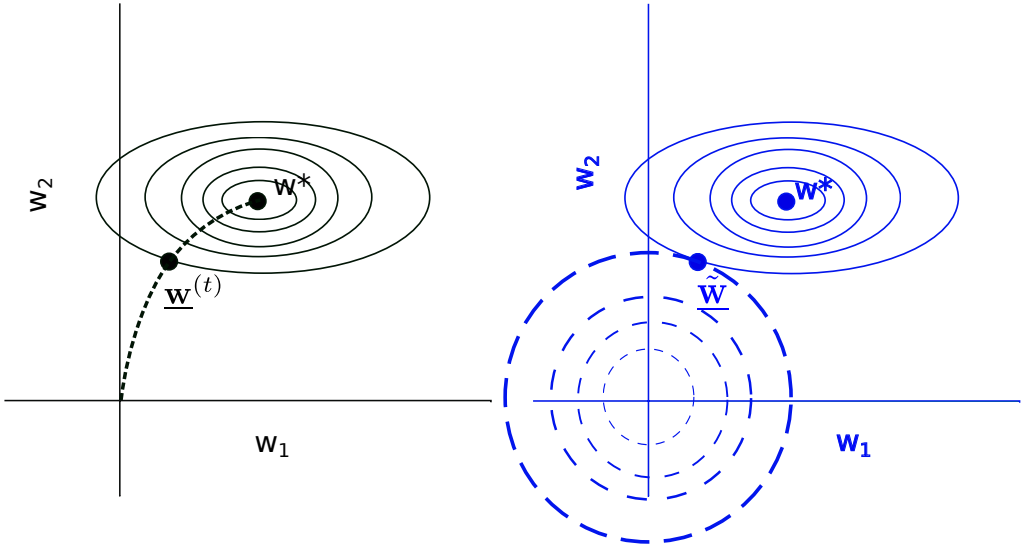
(Deep maxout network trained on MNIST, Goodfellow et al. (2016))

Early stopping as regularization



- Is there a relationship between (unregularized) early stopping at time t and converged *weight-decay* regularization with constant λ?

Early stopping as regularization



- Is there a relationship between (unregularized) early stopping at time t and converged *weight-decay* regularization with constant λ?
- Yes - for parabolic cost functions E^T and initialization with small weights $\underline{\mathbf{w}}^{(0)}$ in the case of weight decay[6].

$$E^R_{(\underline{\mathbf{w}})} = E^T_{(\underline{\mathbf{w}}^*)} + \frac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) + \frac{\lambda}{2}\underline{\mathbf{w}}^\top \underline{\mathbf{w}}$$

$$t \approx \frac{1}{\eta\lambda}, \qquad \lambda \approx \frac{1}{\eta t}$$

[6] e.g. for a linear model with a quadratic cost function

derivation here

Choice of regularization parameter I: model selection and validation

Testset method[7]

1. perform model selection for different values of λ (on training data)
2. select value of λ, which provides best prediction results (on test data); train model for this value of λ on both training and test set.
3. estimate generalization performance of final model (on validation data)

training set	test set	validation set

[7] Model training is done using the regularized cost function, but the unregularized cost is used for testing and validation.

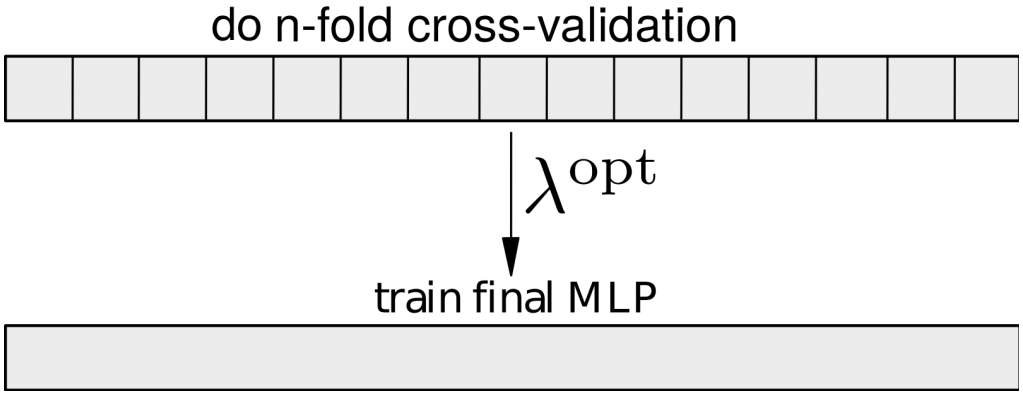
Choice of regularization parameter II: model selection

n-fold cross-validation[8]

for $\lambda = \lambda_1$ **to** λ_m **do**
  perform n-fold cross-validation on all data with regularization parameter λ_i
end

pick optimal λ^{opt} with minimum $\widehat{E}^G$

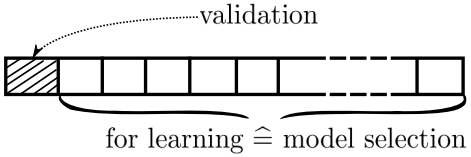
final model: train the model on all data with λ^{opt}



[8] Model training is done using the regularized cost function, but the unregularized cost is used for testing and validation.

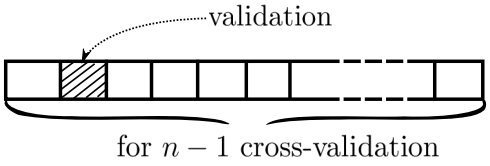
Choice of regularization parameter II: validation

①



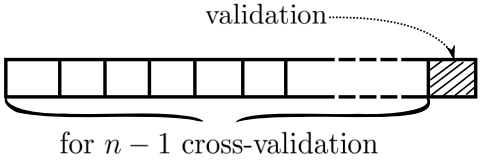
- do n - 1 cross-validation for all candidate values of λ
- train a model with best value for λ
- validate final model

②



- do n - 1 cross-validation for all candidate values of λ
- train a model with best value for λ
- validate final model

ⓝ



- do n - 1 cross-validation for all candidate values of λ
- train a model with best value for λ
- validate final model

- $\widehat{E}^G \quad \widehat{=} \quad$ average over all n validation errors

Choice of regularization parameter II: validation

- Never use test data for model selection (including hyperparameter search).
- Always embed the whole selection procedure (including hyperparameter search) within an n-fold cross-validation run.

1.4.8 Classification Problems (Multi-Class)

Inductive learning

data representation

↓

model class

↓

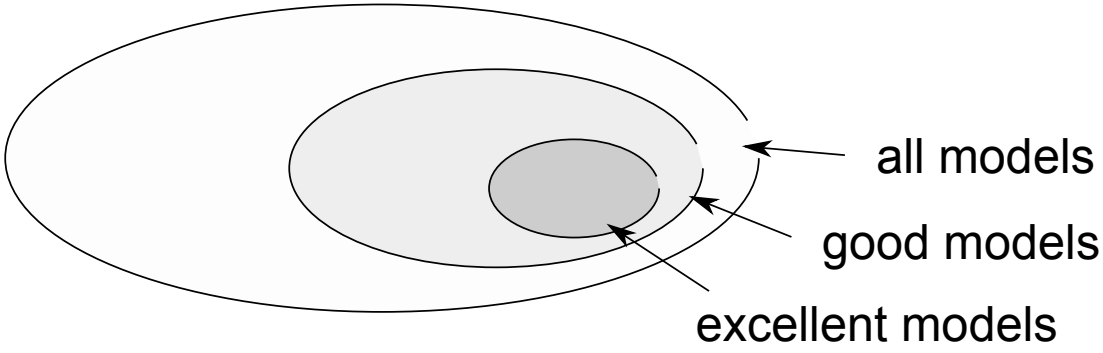
performance measure

↓

optimization

↓

validation



Data representation

Prediction of class labels

observations: $\left\{\left(\underline{\mathbf{x}}^{(\alpha)}, y_T^{(\alpha)}\right)\right\}, \quad \alpha \in \{1, \ldots, p\}$

c classes $C_k, \quad k \in \{1, \ldots, c\} \quad \Rightarrow \quad y_T^{(\alpha)} \in \{1, \ldots, c\}$

Data representation

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c classes $C_k, \quad k \in \{1, \ldots, c\} \quad \Rightarrow \quad y_T^{(\alpha)} \in \{1, \ldots, c\}$

Prediction of class probabilities

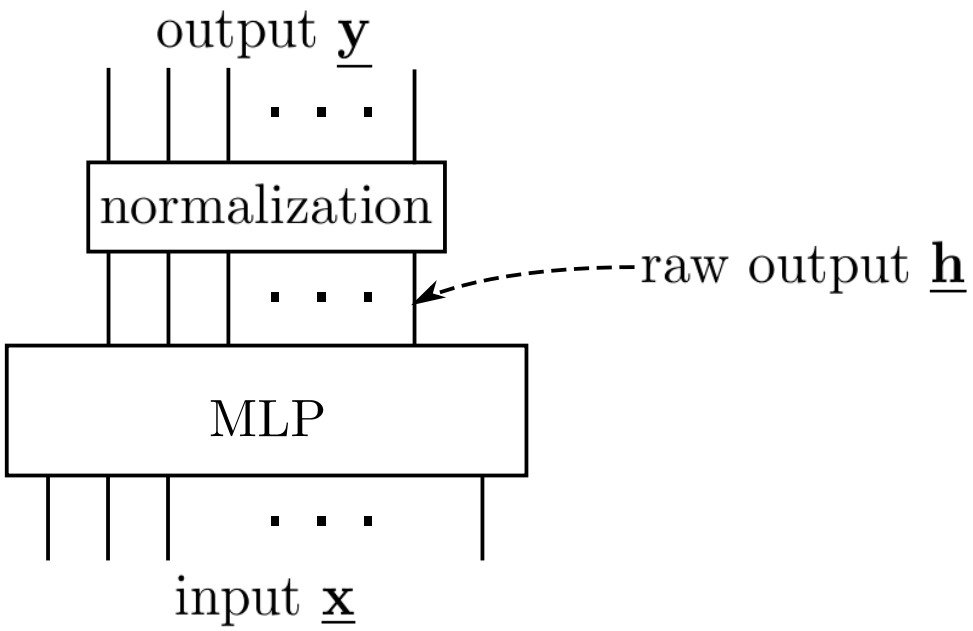
1-out-of-c code

$$y_{Tk}^{(\alpha)} = \begin{cases} 0, & \underline{\mathbf{x}}^{(\alpha)} \notin C_k \\ 1, & \underline{\mathbf{x}}^{(\alpha)} \in C_k \end{cases}$$

$\Rightarrow$ binary vector $\underline{\mathbf{y}}_T^{(\alpha)}$, one non-zero element

Limiting case of probabilities: true labels are known.

Model class



Probabilistic interpretation of network output:

$$y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} := P(\underline{\mathbf{x}} \in C_k; \underline{\mathbf{w}})$$

$$0 \leq \quad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \quad \leq 1$$

$$\sum_{k=1}^{c} y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \quad = 1$$

Softmax normalization

$$y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})})}{\sum_l \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})}$$

Performance measure

true probability: $P_{(C_k|\underline{\mathbf{x}})}$ $\leftrightarrow$ model prediction: $P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}$

Kullback-Leibler-Divergence D_{KL}

$$D_{KL}\left(P_{(C_k,\underline{\mathbf{x}})}\middle\|P_{(C_k,\underline{\mathbf{x}}|\underline{\mathbf{w}})}\right)=\sum_{k=1}^{c}\int d\underline{\mathbf{x}}\,\overbrace{P_{(\underline{\mathbf{x}})}P_{(C_k|\underline{\mathbf{x}})}}^{P_{(C_k,\underline{\mathbf{x}})}}\ln\underbrace{\left(\overbrace{\frac{P_{(C_k|\underline{\mathbf{x}})}}{P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}}\cdot\frac{P_{(\underline{\mathbf{x}})}}{P_{(\underline{\mathbf{x}})}}}^{P_{(C_k,\underline{\mathbf{x}})}}\right)}_{P_{(C_k,\underline{\mathbf{x}}|\underline{\mathbf{w}})}}$$

- distance measure between probability distributions
- non-negative, $D_{KL}=0$ iff $P_{(C_k,\underline{\mathbf{x}})}\equiv P_{(C_k,\underline{\mathbf{x}}|\underline{\mathbf{w}})}$ (distributions are equal)
- asymmetric, $D_{KL}(p\|q)$ does not generally equal $D_{KL}(q\|p)$

Performance measure

true probability: $P_{(C_k|\underline{\mathbf{x}})}$ $\leftrightarrow$ model prediction: $P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}$

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- distance measure between probability distributions
- non-negative, $D_{KL}=0$ iff $P_{(C_k,\underline{\mathbf{x}})}\equiv P_{(C_k,\underline{\mathbf{x}}|\underline{\mathbf{w}})}$ (distributions are equal)
- asymmetric, $D_{KL}(p\|q)$ does not generally equal $D_{KL}(q\|p)$

$$D_{KL}=-\sum_{k=1}^{c}\int d\underline{\mathbf{x}}P_{(\underline{\mathbf{x}})}P_{(C_k|\underline{\mathbf{x}})}\ln P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}+\underbrace{\sum_{k=1}^{c}\int d\underline{\mathbf{x}}\,P_{(\underline{\mathbf{x}})}P_{(C_k|\underline{\mathbf{x}})}\ln P_{(C_k|\underline{\mathbf{x}})}}_{\text{independent of model parameters}}$$

Cross entropy

$$E^G \equiv -\sum_{k=1}^{c} \int d\underline{\mathbf{x}} \underbrace{P_{(\underline{\mathbf{x}})} P_{(C_k|\underline{\mathbf{x}})}}_{\text{unknown!}} \ln P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}$$

- mathematical expectation $\rightarrow$ empirical average over the training set:

$$E^T = -\frac{1}{p} \sum_{\alpha=1}^{p} \ln P(\underline{\mathbf{y}}_T^{(\alpha)} | \underline{\mathbf{x}}^{(\alpha)}; \underline{\mathbf{w}})$$

- label vector assigned to a data point of the training set indicating class membership:

$$\underline{\mathbf{y}}_T = (0, 0, \ldots, 1, \ldots, 0)^\top$$

Cross entropy

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- label vector assigned to a data point of the training set indicating class membership:

$$\underline{\mathbf{y}}_T = (0, 0, \ldots, 1, \ldots, 0)^\top$$

- predicted probability that this data point belongs to its true class:

$$P_{(\underline{\mathbf{y}}_T|\underline{\mathbf{x}};\underline{\mathbf{w}})} = \prod_{k=1}^{c} P_{(\underline{\mathbf{x}} \in C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}{}^{y_{Tk}} = \prod_{k=1}^{c} (y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})})^{y_{Tk}} \,,$$

$$y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \geq 0 \,, \quad \forall k, \quad \text{and} \quad \sum_{k=1}^{c} y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = 1 \,, \quad \forall \underline{\mathbf{x}} \,, \forall \underline{\mathbf{w}}$$

- predicted log probability: $\ln P_{(\underline{\mathbf{y}}_T|\underline{\mathbf{x}};\underline{\mathbf{w}})} = \sum_{k=1}^{c} y_{Tk} \ln \left(y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \right)$

Cross entropy

$$E^G \equiv -\sum_{k=1}^{c} \int d\underline{\mathbf{x}} \underbrace{P_{(\underline{\mathbf{x}})} P_{(C_k|\underline{\mathbf{x}})}}_{\text{unknown!}} \ln P_{(C_k|\underline{\mathbf{x}};\underline{\mathbf{w}})}$$

- mathematical expectation $\rightarrow$ empirical average over the training set:

$$E^T = -\frac{1}{p} \sum_{\alpha=1}^{p} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln \left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right) := \frac{1}{p} \sum_{\alpha=1}^{p} e^{(\alpha)}$$

- label vector assigned to a data point of the training set indicating class membership:

$$\underline{\mathbf{y}}_T = (0, 0, \ldots, 1, \ldots, 0)^\top$$

Optimization via gradient descent (on-line)

$$\Delta \underline{\mathbf{w}} = -\eta \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}$$

$$\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} = -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln \left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right) \qquad \text{with} \quad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}$$

$$= \text{see blackboard...}$$

Optimization via gradient descent (on-line)

$$\Delta \underline{\mathbf{w}} = -\eta \frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}}$$

$$\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} = -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln \left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right) \qquad \text{with} \quad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}$$

$$= \sum_{k=1}^{c} \left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} - y_{Tk}^{(\alpha)} \right) \underbrace{\frac{\partial h_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}}}_{\rightsquigarrow \text{backprop}}$$

Validation

- test-set method or n-fold cross-validation
- BUT: using the cross-entropy cost

$$\hat{E}^G = \frac{1}{q}\sum_{\beta=1}^{q} e^{(\beta)} = -\frac{1}{q}\sum_{\beta=1}^{q}\sum_{k=1}^{c} y_{Tk}^{(\beta)} \ln\left(y_{k(\underline{\mathbf{x}}^{(\beta)};\underline{\mathbf{w}})}\right)$$

- This is fine for “just" assessing overfitting, however, the cross-entropy cost is difficult to interpret.
- How can we convert this to a measure of classification accuracy?

Prediction of class labels

- Decision costs:

  $\$_{ij}$: cost for choosing C_i when $\underline{\mathbf{x}}$ is of class C_j

	patient is sick	patient is not sick
prediction: sick	buy medicine ($20)	adverse effects ($100)
prediction: not sick	sick leave ($1500)	everything is fine ($0)

- Choose C_k with minimal prediction costs:

  $$k = \underset{i}{\operatorname{argmin}} \sum_{j=1}^{c} \$_{ij} \; y_j(\underline{\mathbf{x}}; \underline{\mathbf{w}})$$

- If all errors are equally costly, choose class k which has the highest probability.
- If the prediction of class labels is the final goal, you may use a metric based on "fraction correct" for model validation and hyperparameter selection.

End of Section 1.4

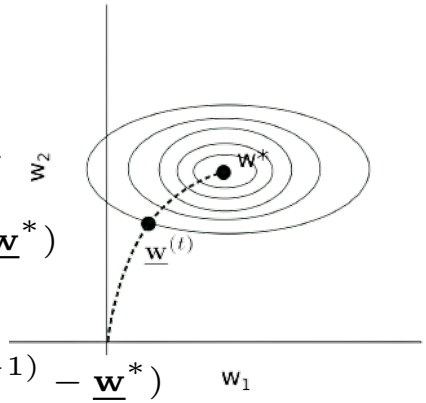
the following slides contain

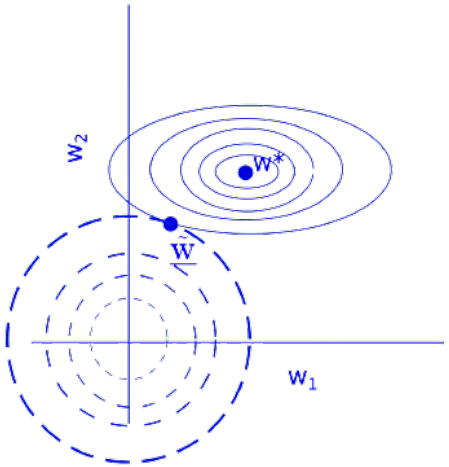
OPTIONAL MATERIAL

Early stopping as regularization

$$E^R_{(\underline{\mathbf{w}})} \quad \approx \quad E^T_{(\underline{\mathbf{w}}^*)} + \tfrac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)$$

$$\begin{aligned}
&\underline{\mathbf{w}}^{(t)} \\
&= \underline{\mathbf{w}}^{(t-1)} - \eta \frac{\partial E^T(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} \\
&= \underline{\mathbf{w}}^{(t-1)} - \eta \underline{\mathbf{H}}(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)
\end{aligned}$$





$$\underline{\mathbf{w}}^{(t)} - \underline{\mathbf{w}}^* = (\underline{\mathbf{I}} - \eta \underline{\mathbf{H}})(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)$$

$$\underline{\mathbf{w}}^{(t)} - \underline{\mathbf{w}}^* = -(\underline{\mathbf{I}} - \eta \underline{\mathbf{H}})^t \underline{\mathbf{w}}^*, \quad \text{initial values } \underline{\mathbf{w}}^{(0)} \approx \underline{\mathbf{0}}.$$

Early stopping as regularization

$$E^R_{(\underline{\mathbf{w}})} \quad \approx \quad E^T_{(\underline{\mathbf{w}}^*)} + \tfrac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)$$

$$\begin{aligned}
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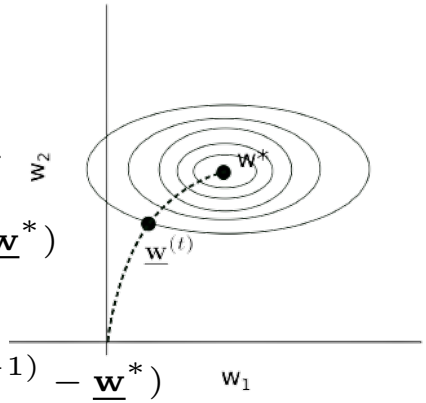
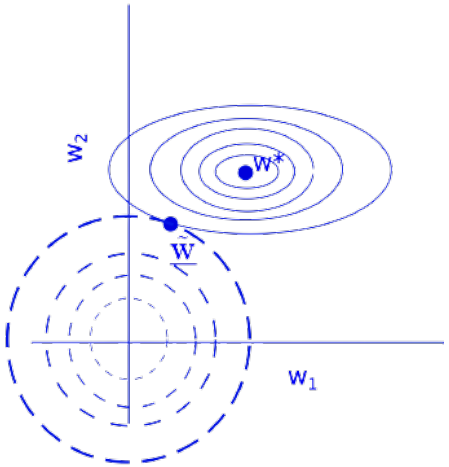
&= \underline{\mathbf{w}}^{(t-1)} - \eta \frac{\partial E^T(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} \\
&= \underline{\mathbf{w}}^{(t-1)} - \eta \underline{\mathbf{H}}(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)
\end{aligned}$$





$$\frac{\partial E^R(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} = \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) + \lambda\, \underline{\mathbf{w}} \stackrel{!}{=} 0$$

$$\Rightarrow \underline{\tilde{\mathbf{w}}} = (\underline{\mathbf{H}} + \lambda\, \underline{\mathbf{I}})^{-1} \underline{\mathbf{H}}\, \underline{\mathbf{w}}^*$$

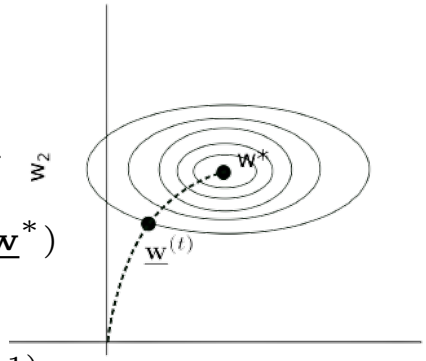
$$\underline{\mathbf{w}}^{(t)} - \underline{\mathbf{w}}^* = (\underline{\mathbf{I}} - \eta \underline{\mathbf{H}})(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)$$

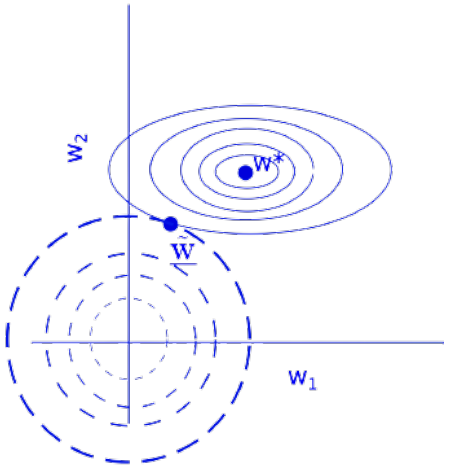
$$\underline{\mathbf{w}}^{(t)} - \underline{\mathbf{w}}^* = -(\underline{\mathbf{I}} - \eta \underline{\mathbf{H}})^t \underline{\mathbf{w}}^*, \quad \text{initial values } \underline{\mathbf{w}}^{(0)} \approx \underline{\mathbf{0}}.$$

Early stopping as regularization

$$E^R_{(\underline{\mathbf{w}})} \quad \approx \quad E^T_{(\underline{\mathbf{w}}^*)} + \tfrac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) \; + \; \tfrac{\lambda}{2}\, \underline{\mathbf{w}}^\top \underline{\mathbf{w}}$$

$$\begin{aligned} &\underline{\mathbf{w}}^{(t)} \\ &= \underline{\mathbf{w}}^{(t-1)} - \eta \frac{\partial E^T(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} \\ &= \underline{\mathbf{w}}^{(t-1)} - \eta \underline{\mathbf{H}}(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*) \end{aligned}$$





$$\frac{\partial E^R(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} = \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) + \lambda\, \underline{\mathbf{w}} \overset{!}{=} 0$$

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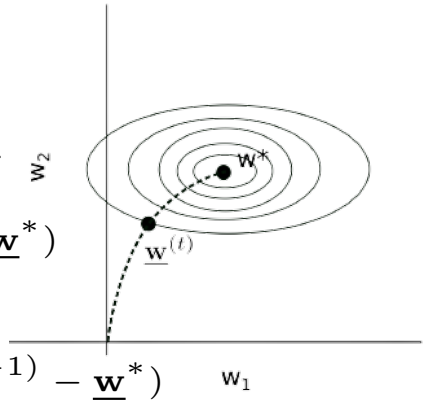
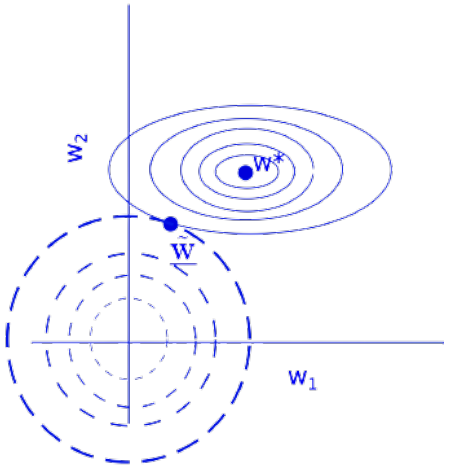
$$\underline{\mathbf{w}}^{(t)} \overset{!}{=} \underline{\tilde{\mathbf{w}}} \quad \Rightarrow \quad \underline{\mathbf{I}} - (\underline{\mathbf{I}} - \eta \underline{\mathbf{H}})^t = (\underline{\mathbf{H}} + \lambda \underline{\mathbf{I}})^{-1} \underline{\mathbf{H}}$$

Early stopping as regularization

$$E^R_{(\underline{\mathbf{w}})} \quad \approx \quad E^T_{(\underline{\mathbf{w}}^*)} + \tfrac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) \; + \; \tfrac{\lambda}{2}\,\underline{\mathbf{w}}^\top \underline{\mathbf{w}}$$

$$\begin{aligned}
&\underline{\mathbf{w}}^{(t)} \\
&= \underline{\mathbf{w}}^{(t-1)} - \eta \frac{\partial E^T(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} \\
&= \underline{\mathbf{w}}^{(t-1)} - \eta \underline{\mathbf{H}}(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)
\end{aligned}$$





$$\frac{\partial E^R(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}} = \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) + \lambda\,\underline{\mathbf{w}} \stackrel{!}{=} 0$$

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$$\underline{\mathbf{H}} =: \underbrace{\underline{\mathbf{U}}\;\underline{\mathbf{K}}\;\underline{\mathbf{U}}}_{\text{diagonal}}{}^\top \quad \Rightarrow \quad \underline{\mathbf{U}}\Big(\underbrace{\underline{\mathbf{I}} - (\underline{\mathbf{I}} - \eta\underline{\mathbf{K}})^t}_{\text{diagonal}}\Big)\underline{\mathbf{U}}^\top = \underline{\mathbf{U}}(\underbrace{\underline{\mathbf{K}} + \lambda\underline{\mathbf{I}})^{-1}\underline{\mathbf{K}}}_{\text{diagonal}}\,\underline{\mathbf{U}}^\top$$

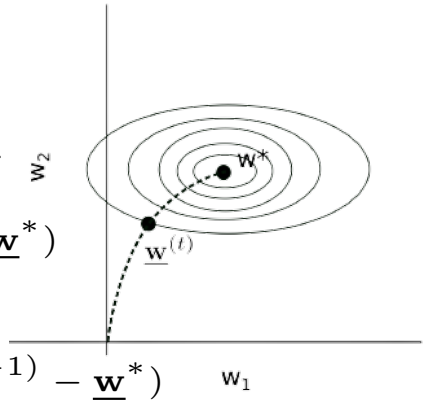
Early stopping as regularization

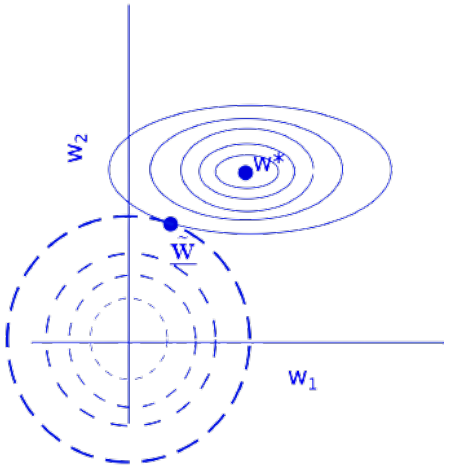
$$E^R_{(\underline{\mathbf{w}})} \quad \approx \quad E^T_{(\underline{\mathbf{w}}^*)} + \tfrac{1}{2}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*)^\top \underline{\mathbf{H}}(\underline{\mathbf{w}} - \underline{\mathbf{w}}^*) \; + \; \tfrac{\lambda}{2}\, \underline{\mathbf{w}}^\top \underline{\mathbf{w}}$$

$$\underline{\mathbf{w}}^{(t)}$$

$$= \underline{\mathbf{w}}^{(t-1)} - \eta \frac{\partial E^T(\underline{\mathbf{w}}^{(t-1)})}{\partial \underline{\mathbf{w}}}$$

$$= \underline{\mathbf{w}}^{(t-1)} - \eta \underline{\mathbf{H}}(\underline{\mathbf{w}}^{(t-1)} - \underline{\mathbf{w}}^*)$$





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$$1 - (1 - \eta \kappa_i)^t = \frac{\kappa_i}{\kappa_i + \lambda} = 1 - \frac{\lambda}{\kappa_i + \lambda}$$

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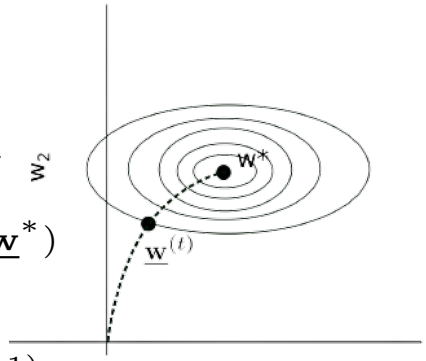

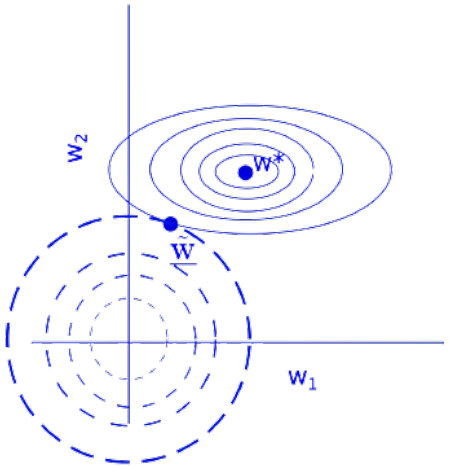

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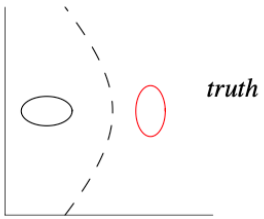
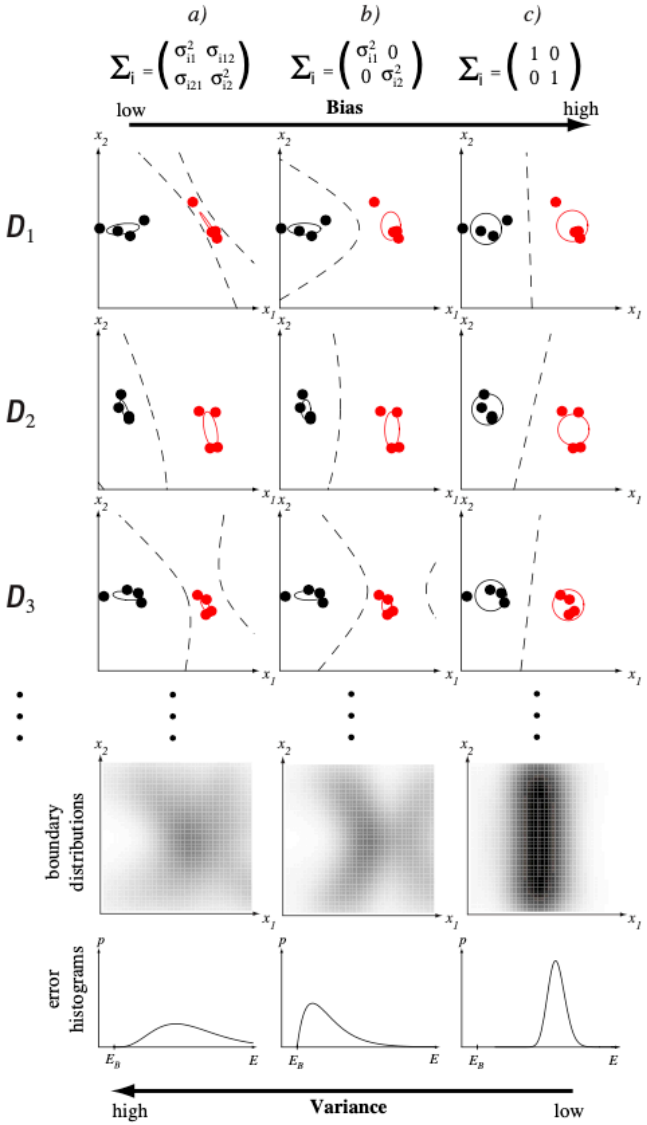
$$1 - (1 - \eta \kappa_i)^t = \frac{\kappa_i}{\kappa_i + \lambda} = 1 - \frac{\lambda}{\kappa_i + \lambda}$$

Taking the logarithm, using the series expansion for $\log(1+x)$, and assuming $\eta \kappa_i << 1$ and $\kappa_1 / \lambda << 1$ we obtain:

$$t \approx \frac{1}{\eta(\kappa_i + \lambda)} \approx \frac{1}{\eta\, \lambda}\,, \quad \lambda \approx \frac{1}{\eta\, t}\,.$$

Classification example





- bias-variance trade-off also applies to classification ("boundary error")
- BUT – for classification: not additive & variance dominates

from Hart et al., 2000, Ch 9.4.

Gradient of softmax normalization

Network computation of probabilities

$$y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})})}{\sum_l \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})} \qquad \text{(softmax function)}$$

Gradient of softmax normalization

Network computation of probabilities

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$$\frac{\partial y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} = \frac{\exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}) \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}) - \exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}) \frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})}{\left(\sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})\right)^2}$$

Gradient of softmax normalization

Network computation of probabilities

$$y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})})}{\sum_l \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})} \qquad \text{(softmax function)}$$

$$\begin{aligned}
\frac{\partial y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} &= \frac{\exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}) \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}) - \exp(h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}) \frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})}{\left(\sum_{l=1}^{c} \exp(h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})})\right)^2} \\
&= y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} - y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \sum_{l=1}^{c} y_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}}
\end{aligned}$$

Gradient of Cross Entropy

$$\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} = -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln \left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})} \right)$$

Gradient of Cross Entropy

$$\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} = -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln\left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\right) \qquad \text{with} \qquad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}$$

Gradient of Cross Entropy

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$$= -\sum_{k=1}^{c} \frac{y_{Tk}}{y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \cdot \frac{\partial y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}}$$

Gradient of Cross Entropy

$$\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} = -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln\left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\right) \qquad \text{with} \qquad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}$$

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Gradient of Cross Entropy

$$
\begin{aligned}
\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} &= -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln\left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\right) \qquad \text{with} \qquad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \\
&= -\sum_{k=1}^{c} \frac{y_{Tk}}{y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \cdot \frac{\partial y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \\
&= -\sum_{k=1}^{c} \frac{y_{Tk}}{y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \left(y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} - y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \sum_{l=1}^{c} y_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \right) \\
&= -\sum_{k=1}^{c} y_{Tk} \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} + \underbrace{\left(\sum_{k=1}^{c} y_{Tk} \right)}_{=1} \left(\sum_{l=1}^{c} y_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \right)
\end{aligned}
$$

Gradient of Cross Entropy

$$
\begin{aligned}
\frac{\partial e^{(\alpha)}}{\partial \underline{\mathbf{w}}} &= -\frac{\partial}{\partial \underline{\mathbf{w}}} \sum_{k=1}^{c} y_{Tk}^{(\alpha)} \ln\left(y_{k(\underline{\mathbf{x}}^{(\alpha)};\underline{\mathbf{w}})}\right) \qquad \text{with} \qquad y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} = \frac{\exp h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\sum_l \exp h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \\
&= -\sum_{k=1}^{c} \frac{y_{Tk}}{y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \cdot \frac{\partial y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \\
&= -\sum_{k=1}^{c} \frac{y_{Tk}}{y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}} \left(y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} - y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} \sum_{l=1}^{c} y_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})} \frac{\partial h_{l(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}} \right) \\
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&= \sum_{k=1}^{c} \left(y_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})} - y_{Tk} \right) \underbrace{\frac{\partial h_{k(\underline{\mathbf{x}};\underline{\mathbf{w}})}}{\partial \underline{\mathbf{w}}}}_{\rightsquigarrow \text{backprop}}
\end{aligned}
$$

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